

Beliefs and Realities of Work and Childcare After Childbirth^{*}

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Abstract

When women plan for life after childbirth, they form beliefs about work, childcare, and how their careers will unfold. These expectations shape key decisions but are formed under deep uncertainty. We use a 2019 state-contingent survey of 11,000 Danish women linked to administrative data to compare pre-birth beliefs to realized outcomes. Mothers accurately anticipate their eventual return to work but underestimate the duration of the career interruption: at nine months postpartum, first-time mothers overestimate their employment by nearly 20 percentage points, an error larger than the raw employment gap between mothers whose partner does and does not take leave. This miscalibration stems from two belief errors – about partner leave and own labor supply – which interact and show no significant sign of narrowing among second-time mothers, with implications for labor supply, planning, and policy design.

Keywords: Children, employment expectations, administrative data.

JEL classification: D31, D84, E24, J31

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1 Introduction

Imagine a mother-to-be thinking ahead to life after childbirth. She is thinking about working after childbirth, how caregiving will be shared, and whether her career plans will stay on track. Like many others, she forms expectations about what life with a child will look like. These beliefs shape key decisions – when to have children, how much to work, and how to plan financially – yet they are formed under deep uncertainty. To understand gender inequality in the labor market, we must understand not only how women respond to the realities of motherhood but also what they expect in advance, and whether those expectations are well-calibrated.

Despite major gains in female education and labor force participation, women continue to experience lower employment rates and earnings than men. A growing body of research pinpoints motherhood and the unequal division of childcare within couples as the central source of this inequality (Goldin, 2024). Employment penalties following childbirth remain substantial, even in contexts with generous leave and childcare policies, suggesting that structural, cultural, or interpersonal dynamics may limit sustained labor market engagement (Kleven et al., 2025; Cortés and Pan, 2023; Boneva et al., 2026). Yet a critical and understudied dimension of this issue is whether women foresee these challenges. Most economic models of female labor supply¹ assume that prospective mothers have perfect foresight or mean correct expectations and plan accordingly – but recent evidence suggests that many underestimate the labor market consequences of having children (Kuziemko et al., 2020). Understanding whether women accurately anticipate the career interruption of motherhood is crucial: if expectations are misaligned, women may enter parenthood unprepared for its consequences, with implications for labor supply, household dynamics, and policy design. This raises a fundamental empirical challenge: to assess whether expectations are accurate, we must elicit beliefs before childbirth and link them credibly to post-birth outcomes. Doing so requires rich, state-contingent data on beliefs about work and high-frequency outcome measures – resources that are rarely available. As a result, there is still almost no direct evidence on whether women’s employment expectations around childbirth align with reality.

We address this empirical gap by combining a large-scale, state-contingent survey with administrative records. In 2019, we asked a representative sample of 11,000 Danish women to report their expectations about working if they were to have a child in the future, under scenarios that varied by whether their partner would take parental leave – a

¹ See for example, Attanasio et al. (2008); Blundell et al. (2016); Eckstein et al. (2019); Adda et al. (2017).

margin previously studied in the context of Swedish parental-leave reforms ([Ekberg et al., 2013](#)), and which we find to be important in this setting as well. Many of these women went on to have children, allowing us to link pre-birth, scenario-specific expectations to realized outcomes. This design makes it possible to assess belief accuracy by capturing expectations before childbirth and linking them to high-frequency administrative data on both post-birth employment and partner leave. We use this framework to measure misalignment and to decompose belief errors into two distinct components: mispredictions about whether the partner will take leave (state beliefs) and mispredictions about one’s own employment given the partner’s leave status (beliefs about state-contingent actions).

We find that most mothers form accurate expectations about their eventual return to work, particularly from 18 months after childbirth and onward. At the same time, they systematically overestimate how soon that return will occur. The most salient misalignment arises around the nine-month mark, a pivotal transition point when many expect to resume work if their partner takes leave: mothers overestimate their likelihood of working at this point by nearly 20 percentage points for first-time mothers and 18 for second-time mothers – for first-time mothers, an error larger than the raw employment gap between mothers whose partner does and does not take leave. These errors reflect miscalibration along both dimensions: women mispredict whether their partner will take leave, and they overestimate their likelihood of returning to work even when the partner takes leave. The belief errors are not random. Misalignment is strongly patterned: women who expect strong partner involvement exhibit compounding errors, while those expecting little support exhibit offsetting ones. These patterns show little sign of narrowing between first and second births, suggesting that experience alone is not sufficient to reliably correct miscalibrated expectations. These findings provide empirical guidance for how to model beliefs in studies of female labor supply and caregiving decisions. For structural modeling of female labor supply around childbirth, our results suggest that rational expectations is a reasonable approximation away from the nine-month transition, particularly for first-time mothers, whose long-run survey-administrative gaps are statistically indistinguishable from zero; short-run belief errors instead concentrate in the partner’s leave-taking margin and the length of the mother’s own leave, not in whether she eventually returns to work at all.

Link to literature: Our analysis contributes to the growing literature on subjective expectations by documenting that beliefs about the transition to parenthood are, for the most part, closely aligned with subsequent realizations – even in this highly routinized, policy-relevant setting – with a systematic, short-run exception that we trace to its source. Unlike prior work, we decompose misalignment into distinct errors about states and state

contingent responses – offering a sharper lens on the structure of expectation errors. This pattern of close overall alignment stands in contrast to [Kuziemko et al. \(2020\)](#), who find that women in the United States and the United Kingdom underestimate the short-run employment costs of motherhood; we argue the difference likely reflects Denmark’s stable, family-friendly institutional environment – unchanged parental leave rules, near-universal childcare access, and social norms that support combining work and family – which reduces the uncertainty that would potentially otherwise generate such underestimation. Our work complements broader research on the drivers of maternal employment reductions, including perceived returns to maternal labor supply ([Boneva et al., 2026](#)), cultural norms around maternal roles ([Kleven et al., 2025](#)), and non-pecuniary costs of motherhood. It also contributes to the literature on child penalties, with [Cortés and Pan \(2023\)](#) providing a comprehensive overview.

Methodologically, we build on [Briggs et al. \(2023\)](#), a companion paper that uses the same survey to develop a general econometric framework for identifying latent propensities to select into treatment and marginal treatment effects from subjective expectations data. That paper’s empirical application is a validation exercise: it compares survey-elicited expectations for the full sample of respondents to the *historical* labor supply of women who had already given birth *before* the survey, and so cannot speak to whether any individual woman’s own expectations predict her own subsequent behavior. Here, by contrast, we follow only women who had children *after* the survey and link each woman’s stated expectations to her own realized outcomes – a direct, one-to-one ex ante/ex post comparison that also lets us examine, for the first time, how expectation accuracy evolves from a first to a second child. [Gong et al. \(2022\)](#) similarly link stated and realized labor supply, but among U.S. college students followed roughly a decade later around age 28; our design instead targets the transition to parenthood itself, isolating short-run belief errors around childbirth that speak to, without directly testing, the type of mechanism [Kuziemko et al. \(2020\)](#) propose. Our use of a state-contingent survey design linked to high-frequency administrative data is central to this contribution, and addresses recent calls for richer belief data ([Almås et al., 2024](#); [Caplin, 2025](#)) by combining subjective expectations, hypothetical scenarios, and objective records to improve our understanding of economic behavior. Leveraging Denmark’s unique institutional and data environment, we show how this design can disentangle the sources of belief bias and reveal systematic heterogeneity in how expectations are formed and biased.

The rest of the paper proceeds as follows. Section 2 describes the institutional context and the data, including the design of the survey and the linked administrative records. Section 3 outlines our empirical strategy for comparing expected and realized employment out-

comes. Section 4 presents the main results, documenting the systematic overestimation at nine months postpartum and decomposing belief errors into state and state-contingent components. We also examine how belief errors vary across groups and evolve with experience across births. Section 5 concludes.

2 Institutional Context and Data

2.1 Child Care and Parental Leave in Denmark

In Denmark the labor participation rate of working-age females is about 80 percent. This is facilitated by a well developed and heavily subsidized high-quality public childcare system and maternity leave policies that protect employment. Public day care is organized by municipalities and it is the dominant type of day care provided. All children are eligible for day care from 26 weeks after childbirth, except if a parent is on publicly funded maternity or parental leave. In practice, since parents are typically on leave until close to the end of the first year after birth, most children do not start day care until around 9–10 months of age, even though formal eligibility begins earlier. Municipalities are legally required to offer a place within a few weeks of a parent’s request, and parents may apply for enrollment at any point during the year, with no seasonal restriction on entry. Opening hours for public day care are typically 6:30am to 5:00pm on weekdays, thus covering the entire work day throughout the work week for most people. Public day care is regulated by law and the objective is to ensure childcare availability for all children and to secure child development and learning. To ensure that public day care in Denmark is of high quality, there are rules that regulate the type and quantity of staffing, physical surroundings, safety, and hygienic standards. Furthermore, municipalities are required by law to monitor childcare institutions to ensure that the personnel have sufficient qualifications and that safety and hygienic standards are met. An important dimension of quality is the staffing. In nurseries (0–2 years) there are typically three children per employee and in kindergartens (3–5 years) there are typically six children per employee and about half of the employees have formal pedagogical training. Public day care is heavily subsidized such that users typically pay about one third of the total cost. The typical monthly user cost for a nursery slot is about 3,000 DKK (1USD $\approx$ 7 DKK) and 1,700 DKK for a kindergarten slot. The design of the public day care system has been in place since the 1970s, which means that childcare opportunities are well known and predictable. 83 percent of all children aged 0–5 years are enrolled in public or publicly funded day care. In other words, day care is not a binding constraint in Denmark.

Danish parents are entitled to up to 52 weeks of leave in connection with having a child. 18 weeks, starting four weeks before childbirth, are earmarked to the mother and two weeks are earmarked to the partner during the period we study. The four weeks earmarked to the mother before childbirth are not fungible: if unused before delivery, they do not extend the leave available after birth. The remaining 32 weeks can be shared between parents as they prefer, and can in principle be taken simultaneously by both parents; in practice, however, most couples take leave sequentially, with fathers using their two earmarked weeks shortly after birth and any additional shared leave taken later, once the mother’s initial leave has ended.² The parental leave scheme guarantees job security and parental leave benefits corresponding to about the level of unemployment insurance benefits. For a full time employee this amounts to about 20,000 DKK per month before taxes in 2020. These rules have been in place since 2002.³ We refer to [Andersen \(2018\)](#) for a more detailed description of the history of the parental leave system.

Using administrative data on parental-leave-related transfer income for the same sample of mothers analyzed in this paper, we find that households make extensive use of the available leave in practice. The median total leave taken by a household is approximately 13 months for the first child and 12 months for the second – close to, and for the first child modestly exceeding, the standard 52-week (approximately 12-month) entitlement, consistent with parents’ option to extend leave by several further weeks without additional paid benefits.⁴ Mothers take the majority of this leave, with a median share of about 81 percent for the first child and 83 percent for the second, while fathers take a median of about 3 and 2 months, respectively. In absolute terms, the median mother is herself on parental leave for 11 months after a first birth and 10 months after a second – a

² In the period we study, mothers were required to inform their employer about the expected due date and planned leave no later than three months before the due date. Partners had to notify their employer about planned paternity leave no later than four weeks before taking leave.

³ In August 2022, parental leave entitlements were restructured to promote a more gender-equal division of leave, reserving 11 additional weeks specifically for fathers. Our survey elicited beliefs in 2019, under the rules described above, and a majority of births in our sample occurred before the August 2022 change. Because we define partner leave-taking as at least two months throughout the paper, a natural concern is that the reform could mechanically push late-cohort partners over this threshold, independent of any change in underlying behavior or beliefs – particularly for the positive state belief error we document among mothers with low $\hat{p}$ (Table 1). Online Appendix D.2 addresses this directly by re-estimating our main results on the subsample of births before the reform: the low- $\hat{p}$ state belief error is essentially unchanged (if anything marginally larger) once the reform-affected cohorts are excluded entirely, and all other results are similar in sign, magnitude, and significance across the two samples.

⁴ Each parent has the right to extend parental leave by eight weeks. Working parents can choose to extend the leave by up to 14 weeks. Extending parental leave does not make either parent eligible for additional parental leave benefits.

figure we return to in Section 4.1 when interpreting the nine-month belief error. There is considerable heterogeneity in leave duration, particularly among fathers, with a minority taking substantially longer leaves than the typical two to three months.

2.2 Data Collection in Denmark

2.2.1 The Custom Survey

The core survey instruments collect probabilities of working at horizons $s = -12, -3, 3, 9, 18$, and 36 months after having the child. Following each question, respondents are shown a slider with a random initial starting point. As respondents slide left and right to choose the probability, the slider updates the selected number so respondents can see their selected percent chance.

We ask respondents to report their beliefs contingent on being in certain states. Because our outcome variable, female employment at different horizons, is Bernoulli, collecting the subjective state-contingent beliefs about outcomes entails collecting the probability of working at different horizons if respondents have a child. We utilize conditional probabilities to collect these distributions:

- “Suppose that you have a child in the next 4 years. What is the percent chance that your partner will take parental leave of at least two months in the first year after the child is born?” We denote this $\hat{p}_i$.⁵
- “Suppose you have a child in the next 4 years and your partner **does** take at least two months parental leave. In this scenario, what is the percent chance you will be working exactly $[[12/3]$ months before/ $[3/9/18/36]$ months after] your child is born?” We denote this subjective probability of working contingent on partner taking leave $\widehat{W}_{ik}^1$ where $k \in \{-12, -3, 3, 9, 18, 36\}$.
- “Suppose you have a child in the next 4 years and your partner **does not** take at least two months parental leave. In this scenario, what is the percent chance you will be working exactly $[[12/3]$ months before/ $[3/9/18/36]$ months after] your child is born?” We denote this subjective probability of working contingent on the partner not taking leave $\widehat{W}_{ik}^0$ where $k \in \{-12, -3, 3, 9, 18, 36\}$.

Respondents report their answers to the second and third questions listed above in a set of stacked sliders, where each line corresponds to a different horizon.

⁵ Throughout we use hat notation to indicate expectations.

Note that the questions are framed explicitly about employment effects of the next child. This means that respondents are asked to imagine the employment effect of having one more child, while disregarding the effects of any additional children beyond that one.

Finally, we ask about age, education, and whether the mother of the respondent was not working, working part-time, or working full-time when the respondent was a child. We also ask about the level of earnings in 2018, Lusardi and Mitchell’s ‘Big Five’ financial literacy question battery⁶, and gender attitudes using the inventory from The International Social Survey Programme.⁷ These questions are listed in the appendix.

2.2.2 Sampling and Link to Registry Data

The sample invited to participate in the Danish survey is recruited from the population registry. The population registry is a complete registry of all persons who are born or have ever had a registered address in Denmark. It contains the CPR-number, which is a personal identifier applied universally to record any contact an individual has with the public sector. For the purpose of conducting our survey we have been given access to a random sample of women aged 18–40 on December 31, 2018. The survey was fielded in May 2019. Invitations to participate were sent out using an official email account called *e-Boks*, which all Danes are equipped with. 51,000 emails were sent out, and we collected 11,312 responses. Participants were not directly paid for completing the survey, but were automatically entered into a lottery among all respondents who completed the full questionnaire: the lottery awarded 50 gift cards worth 1,000 DKK each and one gift card worth 10,000 DKK. This incentive structure was stated in the invitation letter.

In addition to the survey data we also use third-party reported administrative register data compiled by Statistics Denmark from various government agencies. In particular, we use the registries that document household composition, including how many children women have and when they were born. We also rely heavily on data from an administrative register, called the *e-income* registry, that contains employer-reported monthly wages and salaries for employees. All firms in Denmark have to report earnings and hours supplied for each employee, and hence whether the person is working, to the tax agency (SKAT). Since tax evasion on wage income is known to be very limited (Kleven et al. (2011)), we believe this data resource is accurate and precise. This registry has been in place since January 2008 and has previously been used for research purposes by Kreiner et al. (2016) among others. This information is currently available up to and including

⁶ <https://gflec.org/education/questions-that-indicate-financial-literacy/>

⁷ <https://issp.org>.

March 2024. We complement this with information about the receipt of public transfer income and with information about household composition and various background characteristics. Registry data are available for the entire population, and for the purposes of this study, the administrative data have been merged with all women who received invitations to participate. In Table A1 in the Online Appendix we display summary statistics for participants and non-participants. The Table shows that participants are older, more likely to have a partner and to have kids, be Danish citizens, and have higher income and longer educations than non-participants; these differences are often statistically significant, and in relative terms not all of them are small – participants have about 23% higher income and are about 36% more likely to hold a college degree than non-participants. Because our analysis compares each respondent’s own stated expectations to her own realized outcomes, rather than relying on survey participants being representative of the general population in levels, these differences in observable characteristics do not directly threaten our identification strategy; consistent with this, our heterogeneity analysis (Appendix Tables A5 and A6) finds no significant differences in belief accuracy by education or earnings within the sample we do observe.

Our main analysis is further restricted to survey participants who went on to have a child during the observation window (May 2019–April 2023), so it is also useful to compare this group to participants who did not have a child during this period; Appendix Table A2 reports this comparison. Participants who had a child are far more likely to have a partner (77% vs. 56%) and have fewer existing children on average, reflecting where they are in the process of family formation; they also have somewhat higher income (about 15% higher) and are more likely to hold a college degree (31% vs. 21%) than participants who did not have a child in this window. This restriction is not a choice we could relax: our central comparison of stated pre-birth expectations to realized post-birth outcomes is only defined for women who actually gave birth during the period we observe, so the analysis sample necessarily reflects women further along in family formation at the time of the survey. A related concern is selection into birth conditional on fertility intentions: whether belief accuracy differs between women who were, at the time of the survey, more or less certain they would go on to have a(nother) child. Online Appendix C.6 tests this directly using the stated probability of having a(nother) child elicited by Briggs et al. (2023). As with the participant/non-participant comparison above, these differences do not directly threaten our identification strategy, and our heterogeneity analysis finds no significant differences in belief accuracy by education or earnings – two of the dimensions on which this comparison differs most.

3 Estimating Realized Employment around Childbirth

In order to compare realized employment around childbirth with expected employment we compare survey responses to realized employment observed in administrative records for the same women who participated in the survey and for the period which expectations pertain to. Specifically, we compare average employment expectations among women who had children after the survey was collected with average treatment effects on the treated (*ATTs*) using an event study approach with realized labor supply data. To do this we identify our survey respondents who have given birth to children during the period May 2019 to April 2023.

For these childbirths we follow the labor supply of the mother at a monthly frequency for the period covering 24 months before and up to 36 months after childbirth. In order to quantify the employment response around childbirth we estimate the following regression:⁸

$$W_{it} = \sum_{k \in K} B_{it}^k \delta_k + \sum_a \beta_a \cdot \mathbf{I}[a = \text{Age}_{it}] + \sum_{\tau} \lambda_{\tau} \cdot \mathbf{I}[\tau = YM_{it}] + \gamma X_{it} + u_{it}. \quad (1)$$

W_{it} is a dummy variable taking the value one if individual i is working in month t . B_{it}^k is a dummy for the child being born k periods ago and δ_k measures the effect of the childbirth k periods around birth. The month 12 months before childbirth is omitted, so $k \in K = \{-24, -23, \dots, 36\} \setminus \{-12\}$. Following the child penalty literature (Kleven et al., 2019), we include a full set of (single-year) age fixed effects, $\mathbf{I}[a = \text{Age}_{it}]$, and a full set of calendar year-month fixed effects, $\mathbf{I}[\tau = YM_{it}]$, which absorb underlying life-cycle and time trends, respectively. X_{it} is a vector of dummy variables controlling for the timing of adjacent children that the mother has also given birth to; we include these because many women tend to space the arrival of children within 2–3 years, and to make estimated realized employment effects comparable to the elicited expectations, which only concern the next child, it is important to control for the impact of adjacent children on employment. Equation (1) is estimated on a sample of women who all have

⁸ Equation (1) pools women giving birth in different calendar months into a single two-way-fixed-effects regression, which can be biased under staggered treatment timing if treatment effects are heterogeneous across cohorts (Sun and Abraham, 2021). Appendix C.4 implements a cohort-aggregated event-study estimator following Sun and Abraham (2021) and shows that the resulting profile is quantitatively similar to the pooled estimate reported here.

children but at different points in time, and δ_k can therefore be interpreted as the *ATT* at event time k .

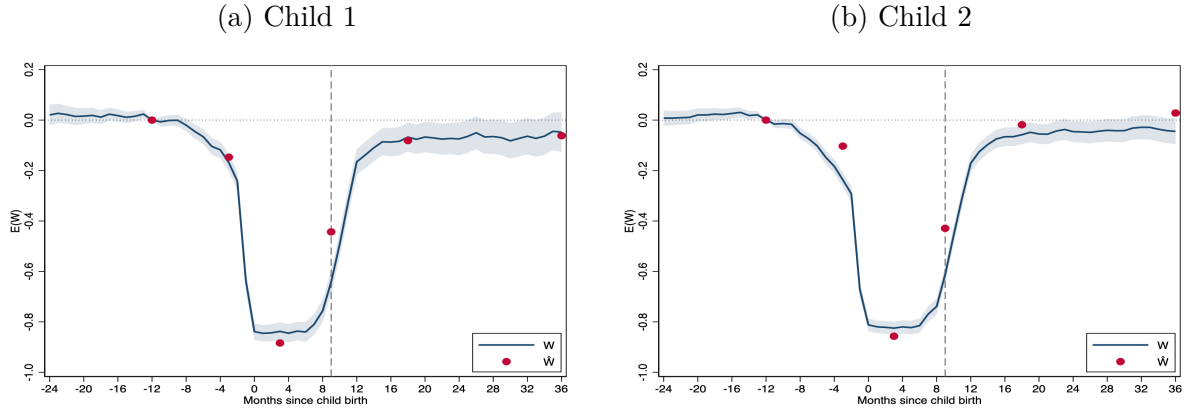
For expectations, we follow the exact same approach, except we only have expectations data pertaining to months -12, -3, 3, 9, 18, 36.

4 Results

4.1 How Unconditional Expectations Align with Reality

We compare expected and realized employment for first- and second-time mothers who had a child during the period from May 2019 to April 2023 and who participated in the survey. A total of 1,574 mothers had their first child and 1,501 had their second child during this window.

Figure 1: Expected and realized employment by 1st and 2nd child



Notes: This figure shows average actual employment, W_i , around childbirth estimated according to (1) with a solid line, and average unconditional expected employment, $\widehat{W}_i = \hat{p}_i \times \widehat{W}_i^1 + (1 - \hat{p}_i) \times \widehat{W}_i^0$, with dots. The shaded region shows the 95% confidence interval, computed from standard errors clustered at the individual level, for realized employment, W_i ; confidence intervals for expected employment, $\widehat{W}_i$, are omitted from the figure for visual clarity but are reported at each surveyed horizon in Appendix Table A3. Panel (a) includes first-time mothers. Panel (b) includes second-time mothers. The dashed line marks nine months after childbirth ($k = 9$).

Figure 1 compares average realized employment outcomes, estimated using equation (1), to average unconditional employment expectations. These expectations are calculated as

a weighted average of expectations under the scenarios where the partner does and does not take parental leave, weighted by the mother’s stated probability that the partner will take leave:

$$\widehat{W}_i = \hat{p}_i \times \widehat{W}_i^1 + (1 - \hat{p}_i) \times \widehat{W}_i^0.$$

Figure 1a shows that first-time mothers, on average, form well-aligned expectations. They accurately anticipate ceasing work prior to childbirth, being out of the labor force three months postpartum, and returning to work by 18 and 36 months postpartum. These patterns are likely shaped by Denmark’s long-standing parental leave and childcare policies, which have contributed to predictable employment trajectories. Appendix Table A3 reports the survey-administrative gap and its standard error at each surveyed horizon. For first-time mothers, the gap is statistically indistinguishable from zero at $k = -3, 18, 36$; it is statistically significant at $k = 3$, but small in magnitude (-0.046). At nine months postpartum, by contrast, we observe a pronounced belief error, more than four times larger than at any other horizon: mothers overestimate their likelihood of working by 0.199 (SE = 0.030).

Figure 1b shows that results are similar for second-time mothers, though with somewhat less precisely estimated zeros away from nine months: the gap is significant at the 10% level at every surveyed horizon (Appendix Table A3), and at the 5% level at three of the five horizons ($k = -3, 9, 36$), reflecting the precision afforded by the sample size. Magnitudes remain informative: gaps away from nine months range from 0.032 to 0.133 in absolute value, compared to 0.183 (SE = 0.022) at nine months, which remains the largest belief error. The two horizons flanking birth differ markedly from each other: at three months before birth, employment falls further than mothers anticipated (gap of 0.133, SE = 0.018), a pattern specific to the run-up to a second birth and unrelated to the return-to-work question that is our focus, whereas at three months after birth the gap is far smaller and only marginally distinguishable from zero (-0.032 , SE = 0.017, $p \approx 0.06$). Neither affects our reading of the figure: There is systematic overestimation of return-to-work at nine months.

This pattern is consistent with realized leave-taking behavior. As reported in Section 2.1, the median mother is herself still on parental leave for 11 months after a first birth and 10 months after a second – both past the nine-month horizon at which she expected, on average, to have returned to work. The nine-month belief error we document is therefore not a mystery about labor market frictions or job search: it is consistent with the fact that most mothers have not yet ended their own leave at the point our survey asked them to predict their employment status.

To assess the robustness of our findings, we explore heterogeneity in belief accuracy across a range of observable characteristics. Specifically, we estimate equation (1) separately by education, age, pre-childbirth earnings, gender attitudes, financial literacy, and childhood exposure to maternal employment. We also examine whether belief accuracy differs for women who are still partnered with the same person at childbirth as at the time of the survey. These dimensions are chosen for their potential relevance to planning, information processing, or caregiving expectations.

Across all subgroups, we find no systematic variation in belief errors. Appendix Tables A5 and A6 test this formally at every surveyed horizon: of 60 tests of the belief-error gap across the six dimensions, only one is significant at the 5% level (financial literacy, first-time mothers, three months before birth), in line with chance. For example, contrary to the hypothesis proposed by Kuziemko et al. (2020), we find no difference in belief accuracy by education level, nor by earnings, gender attitudes, or financial literacy more broadly. Even among mothers with stable partnerships, expectation patterns closely mirror the overall results. Realized employment levels do vary modestly by subgroup, most persistently for second-time mothers by age and earnings, but this reflects differences in labor supply rather than in the accuracy of expectations, which is our focus here. Full details and graphical results for these subgroup analyses are presented in the Online Appendix.

Taken together, these subgroup analyses support the main finding: unconditional expectations and realized employment are broadly aligned, except at nine months postpartum, where a persistent and systematic belief error emerges.

4.2 Partners’ Leave and Maternal Employment Expectations and Outcomes

Our survey includes state-contingent questions about maternal employment expectations under two scenarios: (1) the partner takes at least two months of parental leave, and (2) the partner does not. (Henceforth, we refer to these as the partner taking leave or not.) These questions allow us to examine how expectations and outcomes differ depending on anticipated partner involvement.

Figure 2 plots realized employment and corresponding state-contingent expectations for first- and second-time mothers. In the data, 70 percent of fathers take at least two months of parental leave after the first child, and 64 percent do so after the second child. The blue solid line shows the average employment rate among mothers whose partner actually

took leave, W^1 , while the blue dot shows their average belief in that scenario, $\widehat{W}^1$. The red solid line shows employment when the partner did not take leave, W^0 , and the red dot indicates the corresponding belief, $\widehat{W}^0$.

The comparison of realized outcomes (solid lines) reveals that partner leave-taking is strongly associated with when mothers return to work. At nine months postpartum, the employment gap between mothers whose partner did versus did not take leave is 0.145 (SE = 0.067, $p = 0.03$) for first-time mothers and 0.210 (SE = 0.041, $p < 0.001$) for second-time mothers – both statistically significant at conventional levels.

State-contingent beliefs generally align with realized outcomes, except at $k = 9$, where mothers overestimate their likelihood of being back at work. Among first-time mothers, the difference between actual employment when the partner takes leave (W^1) and the corresponding belief ($\widehat{W}^1$) is 0.234 (SE = 0.034). The gap between actual and expected employment when the partner does not take leave is smaller but still sizable and highly significant, at 0.142 (SE = 0.058). Appendix Table A4 reports the gap and its standard error at every surveyed horizon for both scenarios.

Figure 2b shows comparable patterns for second-time mothers. The belief-outcome gap is 0.203 (SE = 0.027) when the partner takes leave, and 0.150 (SE = 0.031) when not.

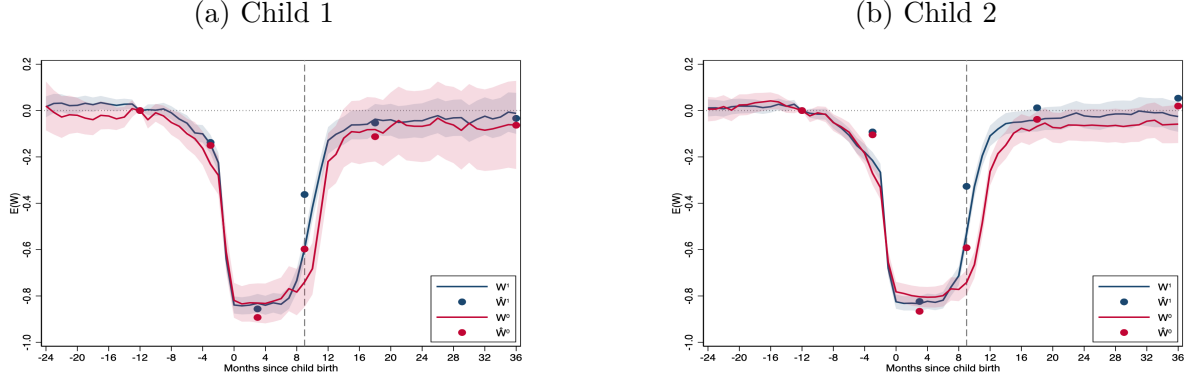
Taken together, these results indicate that mothers’ state-contingent beliefs are generally well-formed – closely tracking observed behavior – except around the nine-month mark. Mothers tend to anticipate their eventual return to work accurately but systematically overestimate how soon it will occur. The magnitude of these belief errors is on par with the actual employment difference between mothers whose partner does and does not take leave.

4.3 Beliefs about Partner’s Parental Leave

The previous section shows that partner leave is strongly associated with women’s beliefs about when they return to work and with when they actually return to work (an association we treat as descriptive throughout, since partner leave-taking is not exogenous – see Section 4.4). Next, we consider women’s beliefs about whether the partner will take leave.

The survey asks mothers to report their expectation of whether their partner will take leave. Using administrative data, we observe whether the partner actually took leave when a child was born. This enables a direct comparison between mothers’ state beliefs

Figure 2: Expected and Realized Employment by Whether Partner Taking Parental Leave



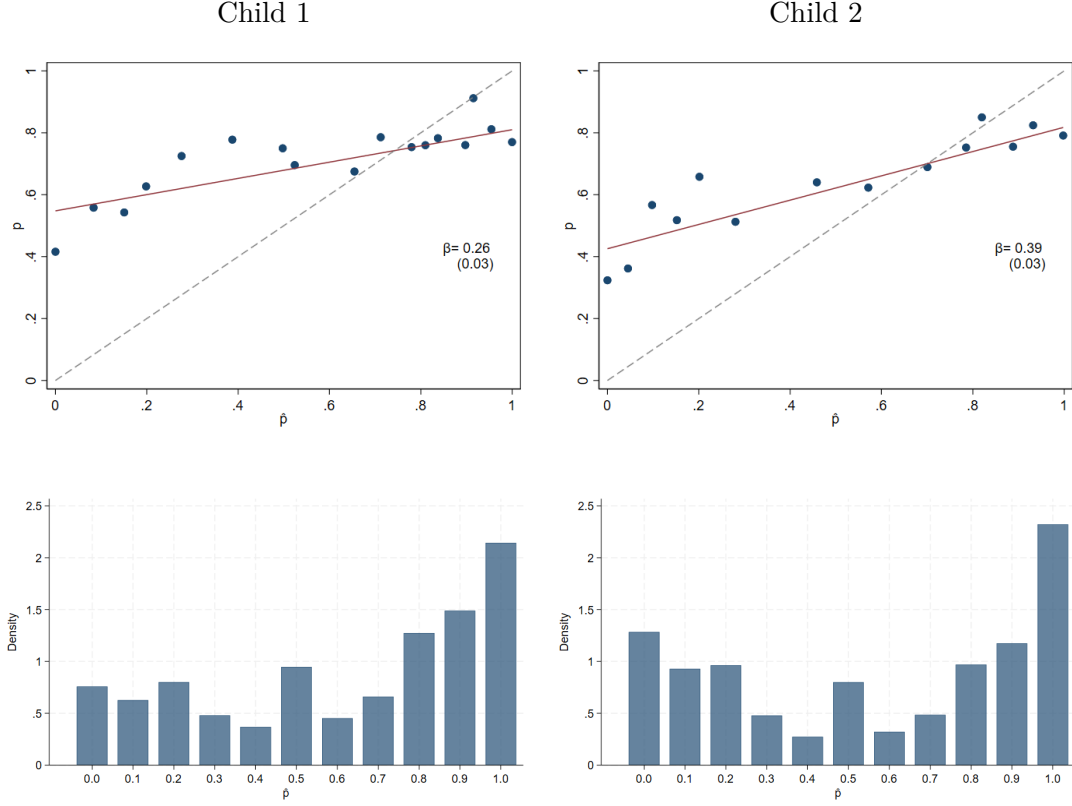
Notes: This figure shows actual employment around childbirth estimated according to (1). Solid lines shows the probability of working for women whose partner actually take leave (W^1) with blue and does not take leave (W^0) with red. State contingent beliefs ($\widehat{W}^1$ and $\widehat{W}^0$) are shown with dots. The shaded regions show the 95% confidence interval, computed from standard errors clustered at the individual level, for realized employment in each scenario; confidence intervals for the state-contingent beliefs are omitted from the figure for visual clarity but are reported at each surveyed horizon in Appendix Table A4. Panel (a) includes first-time mothers. Panel (b) includes second-time mothers. The dashed line marks nine months after childbirth, $k = 9$.

and actual partner behavior.

The upper-left panel of Figure 3 shows the relationship for first-time mothers. The slope of the regression line is 0.26, indicating that mothers' expectations explain only a small share of the variation in actual partner leave. This weak association is consistent with beliefs about partner leave not tracking realized behavior closely, though we cannot observe the planning process directly.

For second-time mothers (upper-right panel), the relationship is slightly stronger, with a slope of 0.39. This may reflect some learning after the first child, though the alignment between beliefs and outcomes remains modest overall. We test formally whether the two slopes differ by pooling first- and second-time mothers into a single regression of realized partner leave on $\hat{p}$, a second-child indicator, and their interaction, clustering standard errors by mother (some mothers contribute to both slope estimates, having appeared as first-time mothers earlier in the sample). The interaction coefficient – the difference between the two slopes – is 0.130 (SE = 0.045, $p = 0.003$), confirming that the slope for second-time mothers is significantly steeper.

Figure 3: Realized Partner Leave by Woman’s Expectation about Partner Leave



Notes: The top row of this figure shows a binned scatter plot of the fraction of partners taking parental leave, p against the expectation of the mother that the partner will take leave, $\hat{p}$, with the fitted regression line (solid) and the 45-degree line of perfect calibration (dashed) shown for reference. Below are shown histograms of $\hat{p}$. The left panels is for mothers giving birth to their first child, the right panels for mothers giving birth to their second child.

The lower panels show the distribution of stated beliefs about partner leave. While mothers report beliefs across the full support from 0 to 1, a substantial share cluster exactly at 0, 0.5, and 1, suggesting some respondents interpret the probability question in a coarse or categorical way rather than reporting a continuously varying probability.⁹ De-

⁹ Among first-time mothers, 7.0%, 7.7%, and 19.6% state $\hat{p}$ equal to exactly 0, 0.5, and 1, respectively (34.3% combined); among second-time mothers, 11.8%, 6.2%, and 21.4% (39.4% combined). Because our results center on average belief errors within a stratum rather than the shape of the belief distribution, we do not expect this to affect our conclusions; future surveys eliciting subjective probabilities could probe such round-number responses with a brief follow-up question (e.g., whether “0”/“100” means “certain” or simply “very unlikely/very likely”).

spite this pattern, the weak correlation between beliefs and outcomes implies meaningful bias in state beliefs – with many mothers either overestimating or underestimating their partner’s likelihood of taking leave.

We also analyze the subsample of mothers who were pregnant at the time of the survey – those for whom leave planning was likely more concrete. In this group, the correlation between expected and realized partner leave is somewhat higher: the slope rises to 0.38 (SE = 0.06) for first-time and 0.46 (SE = 0.05) for second-time mothers. Still, even among this group, the misalignment remains even when decisions are imminent. Online Appendix C.5 replicates our two central empirical exhibits, Figures 3 and 1, on this subsample in full.

The pattern of heterogeneity in state beliefs is particularly important. If beliefs were well-calibrated, the regression lines would lie along the 45-degree line. Instead, we observe that mothers with low $\hat{p}$ tend to underestimate, while those with high $\hat{p}$ tend to overestimate the likelihood that their partner will take leave. These opposing biases offset one another in the aggregate, resulting in an average belief that is close to the average realized probability. This cancellation, however, masks significant underlying belief heterogeneity, reinforcing the need to examine belief errors beyond the average. The fact that there is miscalibration matters because it implies women may systematically underprepare, financially, professionally, or emotionally, for how long they will be out of work after childbirth.

4.4 Decomposing Systematic Belief Errors at $k = 9$

Our results show that future mothers generally have accurate expectations about their return to work from 18 months after childbirth. However, around 9 months postpartum, a clear deviation between beliefs and outcomes emerges: mothers tend to overestimate their likelihood of returning to work at that point. The preceding analysis has documented this misalignment but has not yet disentangled the underlying sources. To address this, we implement a formal decomposition of the belief error in employment expectations observed at nine months after birth.

To facilitate empirical implementation we collect individuals into groups indexed by g . Let $\mathbb{E}_g[\widehat{W}_i^0]$ denote the average stated probability of working if the partner does *not* take leave, and $\mathbb{E}_g[\widehat{W}_i^1]$ the average stated probability of working if the partner *does* take leave for people in group g . Let $\mathbb{E}_g[W_i^0]$ and $\mathbb{E}_g[W_i^1]$ denote the corresponding realized employment probabilities in the data. The average realized employment in group g is given by:

$$\mathbb{E}_g[W_i] = (1 - p_g)\mathbb{E}_g[W_i^0] + p_g\mathbb{E}_g[W_i^1] + \text{Cov}_g(D_i, W_i^1 - W_i^0),$$

where $p_g = \mathbb{E}_g[D_i]$ is the fraction of individuals in group g whose partner actually took leave, and $\text{Cov}_g(D_i, W_i^1 - W_i^0)$ captures within-group sorting on treatment effects.

Similarly, the average subjective belief about employment is:

$$\mathbb{E}_g[\widehat{W}_i] = (1 - \mathbb{E}_g[\widehat{p}_i])\mathbb{E}_g[\widehat{W}_i^0] + \mathbb{E}_g[\widehat{p}_i]\mathbb{E}_g[\widehat{W}_i^1] + \text{Cov}_g(\widehat{p}_i, \widehat{W}_i^1 - \widehat{W}_i^0),$$

where $\mathbb{E}_g[\widehat{p}_i]$ is the average subjective belief that the partner will take leave in group g , and $\text{Cov}_g(\widehat{p}_i, \widehat{W}_i^1 - \widehat{W}_i^0)$ reflects belief-based sorting.

A resulting decomposition of the unconditional belief error in group g is:

$$\begin{aligned} \mathbb{E}_g[W_i] - \mathbb{E}_g[\widehat{W}_i] = & \underbrace{(1 - p_g)(\mathbb{E}_g[W_i^0] - \mathbb{E}_g[\widehat{W}_i^0]) + p_g(\mathbb{E}_g[W_i^1] - \mathbb{E}_g[\widehat{W}_i^1])}_{\text{Error, state-contingent beliefs}} \\ & + \underbrace{(p_g - \mathbb{E}_g[\widehat{p}_i])(\mathbb{E}_g[\widehat{W}_i^1] - \mathbb{E}_g[\widehat{W}_i^0])}_{\text{Error, state beliefs}} \\ & + \underbrace{\text{Cov}_g(D_i, W_i^1 - W_i^0) - \text{Cov}_g(\widehat{p}_i, \widehat{W}_i^1 - \widehat{W}_i^0)}_{\text{Residual sorting terms}} \end{aligned} \quad (2)$$

This decomposition isolates three sources of belief error. The first component reflects errors in state-contingent beliefs – errors in predicting employment for a given partner-leave status. The second component captures errors in state beliefs – miscalibration about whether the partner will take leave. The final line contains residual covariance terms, which capture sorting on treatment effects and expectations.¹⁰

We treat partner leave-taking as a state for analytical convenience, but it is not exogenous in the usual sense: leave-taking is typically the outcome of joint decision-making within the household, and may be correlated with the couple’s anticipated division of caregiving more broadly. Our decomposition does not require partner leave to be uncorrelated with individual-level treatment effects $W_i^1 - W_i^0$: equation (2) already includes a residual sorting term, $\text{Cov}_g(D_i, W_i^1 - W_i^0) - \text{Cov}_g(\widehat{p}_i, \widehat{W}_i^1 - \widehat{W}_i^0)$, that absorbs any such sorting. Empirically, this term’s point estimate is small relative to the other components in every tercile and for both children (Table 1, row (3)); however, these estimates are imprecise so we cannot rule out sorting large enough to matter for the patterns we document. Regardless, belief errors about partner leave-taking should not be interpreted solely as

¹⁰ We refer to the Online Appendix for the derivation of the decomposition.

forecasting mistakes, but potentially also as reflecting the adaptive, jointly-determined nature of household leave decisions.

Table 1: Decomposition of Expectation Bias at $k = 9$, by Levels of $\hat{p}$

	Child 1			Child 2		
	Low	Medium	High	Low	Medium	High
(1) State-contingent belief error						
$(1 - p_g)(\mathbb{E}_g[W_i^0] - \mathbb{E}_g[\widehat{W}_i^0])$	-0.125*** (0.032)	-0.033 (0.036)	0.008 (0.027)	-0.120*** (0.027)	-0.063*** (0.020)	0.000 (0.014)
$p_g(\mathbb{E}_g[W_i^1] - \mathbb{E}_g[\widehat{W}_i^1])$	-0.223*** (0.039)	-0.206*** (0.041)	-0.091** (0.042)	-0.170*** (0.025)	-0.177*** (0.033)	-0.057* (0.031)
Subtotal: State-contingent error	-0.349*** (0.049)	-0.239*** (0.054)	-0.083* (0.050)	-0.290*** (0.035)	-0.241*** (0.038)	-0.057* (0.035)
(2) State belief error						
$(p_g - \mathbb{E}_g[\hat{p}_i])(\mathbb{E}_g[\widehat{W}_i^1] - \mathbb{E}_g[\widehat{W}_i^0])$	0.091*** (0.026)	0.012** (0.005)	-0.056*** (0.009)	0.078*** (0.016)	0.019*** (0.006)	-0.079*** (0.010)
(3) Residual sorting terms						
$\text{cov}(D, W^1 - W^0)$	0.004 (0.077)	-0.005 (0.075)	-0.003 (0.071)	0.003 (0.054)	-0.005 (0.056)	0.023 (0.052)
$\text{cov}(\hat{p}, \widehat{W}^1 - \widehat{W}^0)$	-0.002 (0.002)	-0.000 (0.002)	0.002** (0.001)	-0.001 (0.001)	0.002 (0.003)	0.000 (0.001)
Total bias	-0.252*** (0.056)	-0.232*** (0.052)	-0.143*** (0.049)	-0.207*** (0.039)	-0.229*** (0.041)	-0.114*** (0.037)

Notes: The table presents a decomposition of belief bias according to equation (2). ‘Low’, ‘Medium’, and ‘High’ indicate terciles of the distribution of $\hat{p}$, so each column contains roughly a third of the relevant sample (1,574 first-time and 1,501 second-time mothers in total; Section 4.1). The subtotal under (1) aggregates belief errors conditional on caregiving state; (2) reflects miscalibration about whether the partner will take leave; and (3) includes residual sorting terms, which remain economically small relative to (1) and (2) in every column, even in the one cell where a sorting term is individually statistically significant (0.002, Child 1 High tercile). Underlying regressions for W^1 , W^0 include age and calendar year-month fixed effects (equation (1)). Standard errors in parentheses. Standard errors for the state-contingent subtotal and total bias are obtained by the delta method (via `nlcom`) rather than by combining the component standard errors under an independence assumption. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The individual components of the decomposition, p , $\mathbb{E}[\hat{p}]$, $\mathbb{E}[W^1]$, $\mathbb{E}[W^0]$, $\mathbb{E}[\widehat{W}^1]$, $\mathbb{E}[\widehat{W}^0]$, are tabulated in Online Appendix Table A8.

Table 1 presents the decomposition at 9 months postpartum, stratified by terciles of $\hat{p}$, the stated probability that the partner will take leave. Although the total bias is relatively similar across groups, the decomposition reveals substantial heterogeneity in the sources of error.

Mothers with low $\hat{p}$ systematically underestimate the probability that their partner will take leave (positive state belief error). At the same time, they overestimate their likelihood of working if the partner does not take leave (negative state-contingent belief error). These errors partially offset each other – because they incorrectly assume their partner will not take leave, they anticipate taking on more caregiving, which makes their overall employment expectation more pessimistic, but also unintentionally closer to reality.

By contrast, mothers with high $\hat{p}$ tend to overestimate both the probability of partner leave and their likelihood of working conditional on it. In their case, state and state-contingent belief errors reinforce each other, resulting in similar overall bias levels as the low- $\hat{p}$ group, but driven by different mechanisms.

Throughout Table 1, the state-contingent component is larger in absolute magnitude than the state belief component for most groups. This is not evidence that mothers predict their partner’s behavior better than their own: the two terms are not on a common scale, since the state belief term is scaled by how much expected employment actually differs across partner-leave scenarios (Online Appendix Table A8), a factor well below one throughout, whereas the state-contingent term carries no such scaling. The substantive content of Table 1 lies not in these relative magnitudes but in the pattern established above: whether the two components offset or reinforce one another varies systematically with $\hat{p}$, and it is this pattern of belief formation that the decomposition is designed to reveal.

Table 1’s point estimates also differ somewhat between the first and second child within each tercile, but formally testing these differences (Online Appendix Table A9) shows that eight of the nine comparisons – total bias and each of its two main components, in all three terciles – are not statistically distinguishable from zero, despite point estimates that in places appear directionally consistent with a “learning” story (total bias, for instance, is numerically smaller in magnitude for the second child in all three terciles). The one exception is the state belief error in the high- $\hat{p}$ tercile, which widens rather than narrows from the first to the second child (0.023, SE= 0.014, $p = 0.094$) – significant only at the 10% level, and the opposite of what a simple learning account would predict for this component: it indicates that expectations about partners’ leave-taking do not improve, and if anything worsen, for this group. We therefore do not read Table 1 as strong evidence that belief accuracy systematically improves between births. At most, the high- $\hat{p}$ result is consistent with any improvement being concentrated in mothers’ own state-contingent behavior rather than in their beliefs about their partner, but given how narrowly this single comparison clears conventional significance against eight others that do not, we treat this as suggestive rather than established.

Descriptive characteristics (see Online Appendix Table A7) show that mothers in the upper tercile of $\hat{p}$ tend to be older, more educated, earn more, express less gender-conservative views, and score higher on financial literacy. Their state-contingent beliefs are more accurate, yet they exhibit state belief errors of similar magnitude (though opposite sign) as those in the lower tercile. This composition means the $\hat{p}$ -tercile split partly

proxies for these demographics; however, directly splitting the decomposition by education, earnings, or gender attitudes (Appendix Tables A5 and A6) shows no significant heterogeneity in belief accuracy, so we do not read this pattern as evidence that these observable characteristics themselves drive accuracy – the variation instead appears to run through $\hat{p}$ itself.

Together, these findings underscore the importance of the state-contingent survey design. Without disaggregating beliefs into their component parts, one would overlook meaningful differences in how expectation errors arise. The decomposition demonstrates that similar average levels of belief error can mask distinct patterns of misperception, and that the formation of beliefs around work and childcare varies systematically across groups. It reveals that some women primarily misjudge the distribution of leave-taking within the couple, while others misjudge their own ability to work conditional on the partner’s leave status, and this pattern varies systematically with $\hat{p}$ rather than being random. This distinction matters for both theory and policy, because models that assume homogeneous or mean-correct expectations will miss key behavioral mechanisms, and interventions to improve planning will be more effective when tailored to the specific types of belief errors different women are prone to make.

5 Conclusion

This paper examines how accurately mothers anticipate their return to work after childbirth, using a unique combination of state-contingent survey data and high-frequency administrative records. Our framework distinguishes between two types of belief error: misjudging whether the partner will take leave and misjudging one’s own employment given the partner’s leave status. This distinction helps clarify how expectations about work and childcare are formed – and where they go awry.

We find that while mothers generally anticipate their eventual return to work correctly, they systematically overestimate how quickly they will resume employment – particularly around nine months postpartum, a key moment when leave policies expire and many expect to be back on the job. This misalignment reflects both types of belief error and varies predictably with expectations about partner involvement. Mothers who expect little partner support tend to underestimate actual leave-taking by their partner but also overestimate their own ability to work without it, leading to partially offsetting errors. In contrast, those who expect strong partner involvement tend to misjudge both the partner’s actions and their own, producing compounding errors. These patterns show little sign of narrowing between first and second births, suggestive evidence that expe-

rience alone does not fully correct miscalibrated expectations. These findings highlight belief formation as an important mechanism shaping women’s employment trajectories, alongside the alternative interpretation we discuss next.

Throughout, we have interpreted the systematic overestimation of return-to-work at nine months as a belief error: expectations stated before birth do not match realized behavior. An alternative interpretation is that part of this gap reflects preference discovery rather than a forecasting mistake in the usual sense – mothers may not simply misjudge their eventual return-to-work probability, but revise their preferences over work and caregiving only after experiencing early motherhood, so that a later-than-expected return reflects a deliberate, welfare-improving choice rather than a binding constraint. Denmark’s institutional environment – generous, flexible parental leave and near-universal, heavily subsidized childcare – makes it unlikely that the delayed return we document is driven primarily by financial or administrative constraints, a fact consistent with either interpretation. We think both mechanisms are plausibly at work, and our data cannot fully separate a narrow forecasting error (a systematically wrong point prediction) from preference updating that mothers themselves could not fully anticipate before giving birth. The two interpretations are not mutually exclusive within our framework, however: even if the underlying shift in behavior reflects newly discovered preferences rather than a miscalculation, mothers’ stated expectations do not anticipate this shift, and the resulting gap is concentrated specifically at nine months rather than spread evenly across horizons – both features that still constitute an *ex ante* forecast error in the sense we measure throughout the paper. The distinction nonetheless matters for policy: if the gap reflects a genuine informational or forecasting failure, interventions that help women form more accurate expectations about the likely adjustment path would be called for; if it instead reflects the arrival of new information about caregiving preferences that mothers could not have anticipated before childbirth, policies that preserve flexibility – allowing women to extend leave once such preferences are revealed – may be more appropriate than efforts to accelerate return to work.

Our realized-employment estimates also speak to the broader child-penalty literature. [Kleven et al. \(2019\)](#) document a persistent employment (labor force participation) penalty for Danish mothers of 13 percentage points that remains essentially unchanged up to ten years after a first birth, for cohorts giving birth between 1985 and 2003. Within the shorter, roughly three-year horizon we observe for the more recent cohorts in our sample, employment remains modestly below its pre-birth level even at 36 months (4.5–4.8 percentage points; Appendix Table [A3](#)), though this remaining gap is substantially smaller than the long-run participation penalty documented by [Kleven et al. \(2019\)](#). We

are cautious about over-interpreting this contrast: our data do not extend far enough to say whether a gap similar in size to theirs might still emerge beyond the horizon we observe, and differences in cohorts, sample construction, and outcome definitions may also contribute. Still, the smaller gap we observe within our window is at least suggestive of a fuller near-term recovery of maternal employment under today's more stable and gender-equal parental-leave and childcare environment than was observed for earlier cohorts.

Throughout, we focus on the extensive margin: whether mothers work at all at a given horizon. Misperceptions may also exist on the intensive margin, i.e., expected working hours or the choice between part-time and full-time employment, which our survey does not elicit. Incorporating expectations about hours worked, alongside the extensive-margin measure we study here, is a natural direction for future work.

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Online Appendix

A Summary Statistics and Balance Table

Table A1: Summary statistics for participants and nonparticipants

Variable	(1) Non-participants	(2) Participants	(3) Difference
Age	28.624 (6.606)	29.413 (6.454)	0.789 (0.069)
Partner	0.522 (0.500)	0.607 (0.488)	0.085 (0.005)
Number of Children	0.700 (1.045)	0.782 (1.026)	0.082 (0.011)
Had baby in 2017	0.034 (0.180)	0.039 (0.193)	0.005 (0.002)
Non-DK citizenship	0.190 (0.392)	0.094 (0.292)	-0.096 (0.003)
Income (1000 DKK)	175.536 (16.738)	227.750 (193.482)	52.214 (2.048)
Liquid assets (1000 DKK)	74.988 (484.323)	87.297 (206.467)	12.309 (3.120)
House owner	0.239 (0.426)	0.339 (0.473)	0.100 (0.005)
Student	0.306 (0.461)	0.310 (0.462)	0.004 (0.005)
Educ basic	0.220 (0.415)	0.125 (0.331)	-0.095 (0.004)
Educ high school	0.206 (0.404)	0.184 (0.388)	-0.022 (0.004)
Educ vocational	0.214 (0.410)	0.208 (0.406)	-0.005 (0.004)
Educ middle	0.122 (0.327)	0.190 (0.392)	0.068 (0.004)
Educ College	0.148 (0.355)	0.231 (0.421)	0.083 (0.004)
Observations	39,336	11,312	50,648

Notes: Characteristics are obtained from administrative registries and are measured in 2018. Standard deviations in parentheses in columns (1) and (2). Standard errors in parentheses in column (3).

B Questions

B.1 Earnings in 2018

What was your total salary income in 2018? By this, we mean your gross income, i.e., before taxes are deducted. (Please answer as accurately as possible. It is okay to round off if you do not know the exact amount.)

Please provide your answer in Danish kroner.

B.2 Gender Attitudes

1. A preschool child is likely to suffer if his or her mother works.
2. All in all, family life suffers when the woman has a full-time job.
3. A woman and her family would be happier if she goes to work.
4. Both the man and the woman should contribute to the household income.
5. Having a full-time job is the best way for a woman to be independent.
6. A man's job is to earn money; a woman's job is to take care of the home and family.
7. Children need their father to be as closely involved in their upbringing as their mother.
8. Employers should make a special effort to help mothers combine work and childcare.
9. A single parent is just as good at raising children as a couple.

All the questions have the following answer categories: 'Strongly agree', 'Agree', 'Neither agree nor disagree', 'Disagree', or 'Strongly disagree'. A respondent is coded gender conservative if she answers 'Agree' or 'Strongly Agree' to questions 1, 2, 6, and if she answers 'Disagree' or 'Strongly disagree' to the other questions.

B.3 Financial Literacy Questions

1. Imagine you have 100 DKK in a savings account, and the annual interest rate is 2 percent. How much money do you think will be in the account if you leave it for five years?
 - More than 102 DKK

- Exactly 102 DKK
 - Less than 102 DKK
 - Don't know
 - Prefer not to answer
2. Imagine the interest rate on your savings account is 1% per year, and the inflation rate is 2% per year. If you leave the money in the account for one year, what would you be able to buy?
- More than today
 - Exactly the same as today
 - Less than today
 - Don't know
 - Prefer not to answer
3. Buying a stock in an individual company is a safer investment than investing in a mutual fund.
- True
 - False
 - Don't know
 - Prefer not to answer
4. Imagine you roll a fair six-sided die 1,000 times. Out of those 1,000 rolls, how many times do you think the die will land on an even number?
- [Slider]
5. In a lottery, there is a 1% chance of winning 100,000 DKK. What is your best guess for how many people will win the 100,000 DKK if 1,000 people each buy a lottery ticket?
- 1
 - 10

- 100
- 1000

B.4 Question about Your Mother’s Employment Situation

Please indicate which statement best reflects your mother’s situation when you were under five years old.

- My mother was in full-time employment
- My mother worked part-time
- My mother was a stay-at-home parent
- None of the above

C Additional Analyses

C.1 Statistical Significance of Survey-Administrative Gaps

Appendix Tables [A3](#) and [A4](#) report the exact survey-administrative gap, its standard error, and its statistical significance at every surveyed horizon, underlying Figures [1](#) and [2](#) respectively.

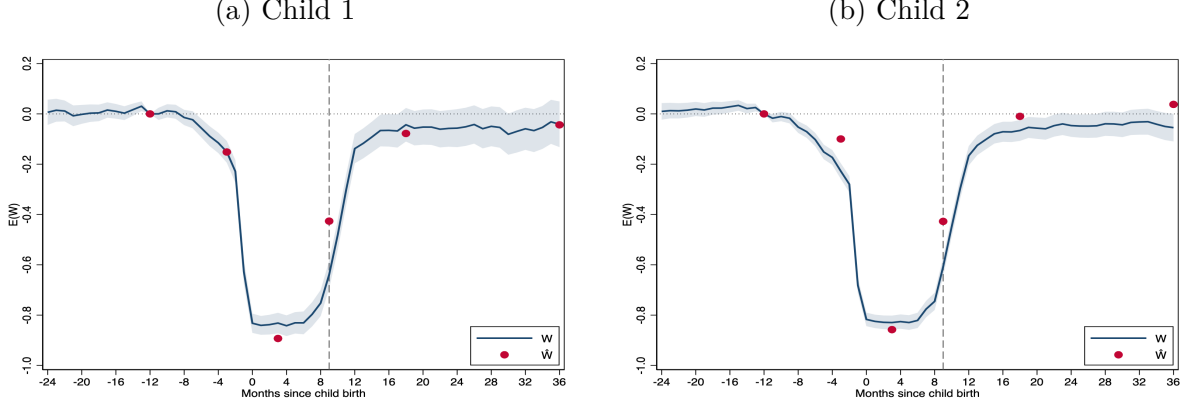
C.2 Conditioning on Having the Same Partner as at Interview

Figure [A1](#) reproduces Figure [1](#), restricted to the subsample of mothers who report having the same partner at the time of the survey as during the estimation window, addressing the possibility that partner turnover affects our results.

C.3 Heterogeneity Along Different Dimensions

We examined heterogeneity along several dimensions, including education, age, pre-childbirth earnings, gender attitudes, financial literacy, and maternal employment during the respondent’s childhood. These analyses found no significant evidence of heterogeneity in the accuracy of expectations – formally tested in Table [A6](#) below, only 1 of 60 tests is significant at the 5% level. Realized employment levels do vary somewhat by subgroup, most persistently for second-time mothers by age and earnings (Table [A5](#)), but this reflects differences in labor supply rather than in the accuracy of expectations.

Figure A1: Expected and realized employment by 1st and 2nd child, same-partner sub-sample



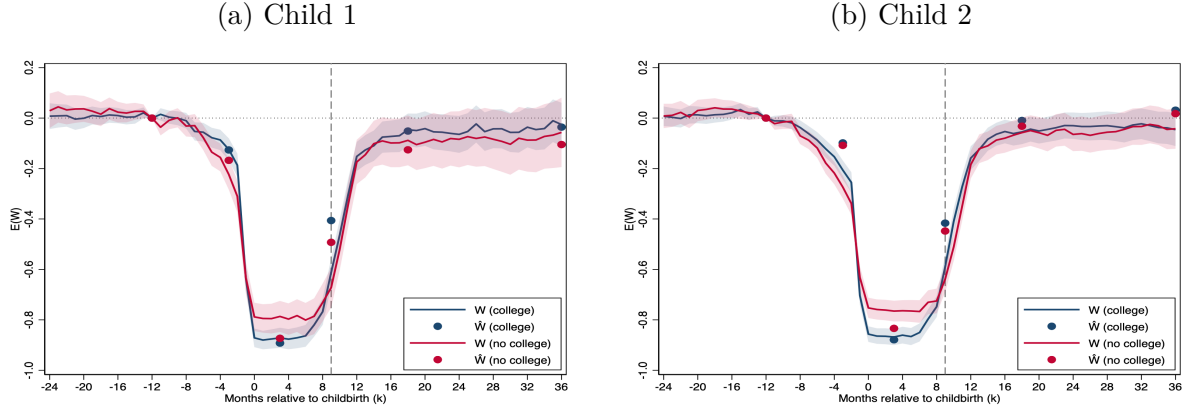
Notes: This figure shows actual employment around childbirth estimated according to (1) with a solid line and unconditional expected employment, $\widehat{W}_i = \hat{p}_i \times \widehat{W}_i^1 + (1 - \hat{p}_i) \times \widehat{W}_i^0$, with dots. The shaded region shows the 95% confidence interval for realized employment, W_i ; confidence intervals for expected employment, $\widehat{W}_i$, are omitted from the figure for visual clarity. The estimations are performed on samples of women who have the same partner as they had at the time of the survey. Panel (a) includes first-time mothers. Panel (b) includes second-time mothers.

Detailed results are presented below: Figure A2 (education), Figure A3 (age), Figure A4 (earnings), Figure A5 (gender attitudes), Figure A6 (financial literacy), and Figure A7 (respondent's own mother's employment).

A natural question is why we stratify the decomposition in Table 1 by $\hat{p}$ rather than by an observable dimension such as education. Splitting the same decomposition by education instead yields the answer: total bias at $k = 9$ ranges by only 0.026 (child 1) and 0.018 (child 2) across college versus non-college mothers, four to six times smaller than the $\hat{p}$ -tercile range of about 0.11 in Table 1, and every individual component moves by a similarly small amount. Education, like the other demographics tested above, does not reveal the kind of pattern $\hat{p}$ does.

Tables A5 and A6 report formal tests corresponding to the six figures above: for each subgroup pair and each surveyed horizon, whether realized employment differs across the subgroups (Table A5), and whether the belief-error gap – the survey-minus-admin over-estimation that is the focus of this paper – differs across the subgroups (Table A6). Since the two subgroups making up each split are non-overlapping partitions of the same

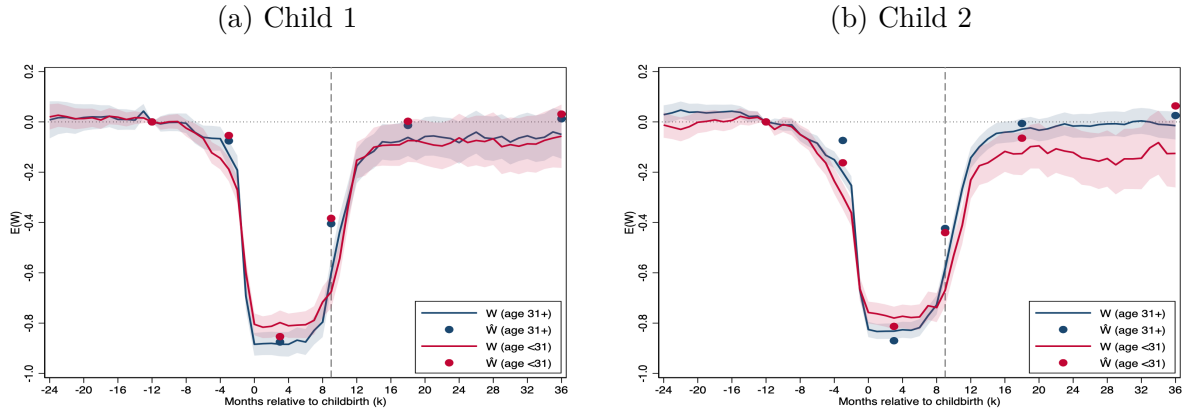
Figure A2: Expected and realized employment by 1st and 2nd child by education



Notes: This figure shows average actual employment, W_i , around childbirth estimated according to (1) and estimated separately for mothers with (blue) and without (red) a college degree. Solid lines are estimated actual employment and dots are average unconditional expected employment, $\widehat{W}_i = \hat{p}_i \times \widehat{W}_i^1 + (1 - \hat{p}_i) \times \widehat{W}_i^0$. The shaded region shows the 95% confidence interval for realized employment, W_i ; confidence intervals for expected employment, $\widehat{W}_i$, are omitted from the figure for visual clarity. Panel (a) includes first-time mothers. Panel (b) includes second-time mothers. Formal tests of whether realized employment and belief-error gaps differ across the subgroups shown in this and the following five figures, at each surveyed horizon, are reported in Appendix Tables A5 and A6.

sample, standard errors for each difference are combined as $\sqrt{se_1^2 + se_2^2}$, exact rather than approximate.

Figure A3: Expected and realized employment by 1st and 2nd child by age



Notes: This figure shows average actual employment, W_i , around childbirth estimated according to (1) and estimated separately for mothers aged at least 31 (blue) and aged less than 31 (red). Solid lines are estimated actual employment and dots are average unconditional expected employment, $\widehat{W}_i = \hat{p}_i \times \widehat{W}_i^1 + (1 - \hat{p}_i) \times \widehat{W}_i^0$. The shaded region shows the 95% confidence interval for realized employment, W_i ; confidence intervals for expected employment, $\widehat{W}_i$, are omitted from the figure for visual clarity. Panel (a) includes first-time mothers. Panel (b) includes second-time mothers.

Table A2: Summary statistics for participants, by whether they had a child during the analysis window

Variable	(1) Did not have children	(2) Had children	(3) Difference
Age	29.685 (6.980)	28.541 (4.245)	-1.144*** (0.111)
Partner	0.556 (0.497)	0.772 (0.420)	0.216*** (0.010)
Number of Children	0.477 (0.733)	0.175 (0.380)	-0.302*** (0.011)
Had baby in 2017	0.033 (0.179)	0.058 (0.234)	0.025*** (0.005)
Immigrant	0.110 (0.312)	0.104 (0.305)	-0.006 (0.007)
Non-DK citizenship	0.095 (0.293)	0.091 (0.288)	-0.003 (0.006)
Income (1000DKK)	219.921 (198.225)	252.816 (175.140)	32.895*** (3.994)
Liquid assets (1000DKK)	85.718 (202.515)	92.353 (218.604)	6.635 (4.744)
House owner	0.339 (0.473)	0.338 (0.473)	-0.001 (0.010)
Student	0.314 (0.464)	0.294 (0.456)	-0.020** (0.010)
Educ basic	0.150 (0.357)	0.052 (0.221)	-0.098*** (0.006)
Educ high school	0.203 (0.402)	0.136 (0.343)	-0.066*** (0.008)
Educ vocational	0.208 (0.406)	0.203 (0.402)	-0.005 (0.009)
Educ middle	0.181 (0.385)	0.212 (0.409)	0.031*** (0.009)
Educ College	0.205 (0.404)	0.312 (0.464)	0.107*** (0.010)
Observations	8,620	2,692	11,312

Notes: Sample restricted to survey participants (Table A1, column (2)). “Had children” indicates participants who gave birth during May 2019–April 2023, the window used for the main analysis; “Did not have children” indicates participants who did not. Characteristics are obtained from administrative registries and are measured in 2018. Standard deviations in parentheses in columns (1) and (2). Standard errors in parentheses in column (3). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Survey-administrative gap in return-to-work expectations, by event-time horizon

k	Survey	Admin. (pooled)	Gap	SE(Gap)	95% CI
<i>Panel A: First child</i>					
-3	-0.147	-0.169	0.021	0.022	[-0.022, 0.065]
3	-0.884	-0.838	-0.046**	0.022	[-0.090, -0.002]
9	-0.443	-0.642	0.199***	0.030	[0.139, 0.258]
18	-0.081	-0.067	-0.014	0.036	[-0.085, 0.057]
36	-0.062	-0.048	-0.014	0.052	[-0.115, 0.087]
<i>Panel B: Second child</i>					
-3	-0.103	-0.237	0.133***	0.018	[0.099, 0.168]
3	-0.857	-0.825	-0.032*	0.017	[-0.065, 0.001]
9	-0.429	-0.612	0.183***	0.022	[0.139, 0.227]
18	-0.018	-0.058	0.040*	0.024	[-0.007, 0.086]
36	0.028	-0.045	0.073**	0.032	[0.011, 0.135]

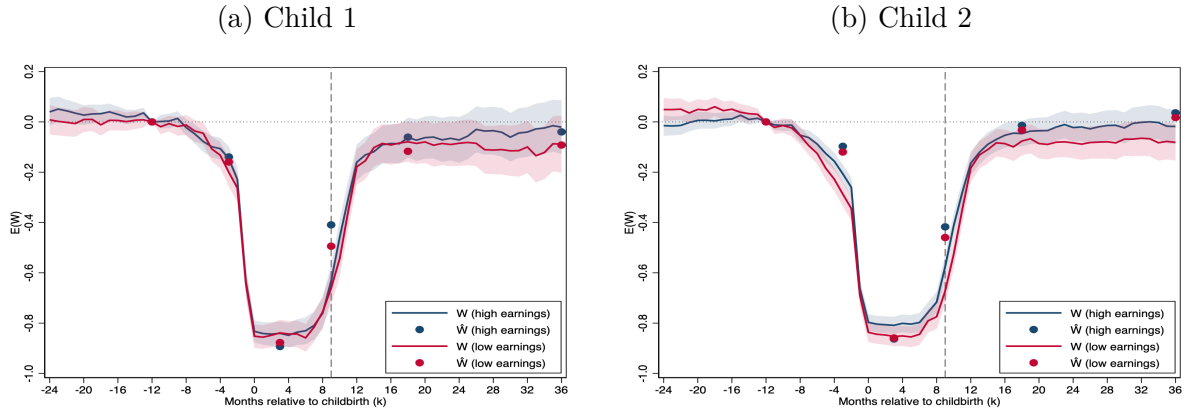
Notes: k denotes months relative to childbirth. Survey and Admin. (pooled) columns report, respectively, the survey-elicited and pooled TWFE event-study estimates of labor-market attachment at horizon k , expressed as deviations from $k = -12$. Admin. estimates include age and calendar year-month fixed effects (equation (1)). Gap = Survey - Admin. $SE(\text{Gap}) = \sqrt{SE(\text{Survey})^2 + SE(\text{Admin})^2}$, which assumes the two estimates are independent; since both are estimated on the same women, this is a conservative (slightly too wide) approximation rather than the exact joint SE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, two-sided z -test of Gap = 0.

Table A4: Survey-administrative gap in state-contingent employment expectations, by partner-leave scenario

k	Survey	Admin. (pooled)	Gap	SE(Gap)	95% CI
<i>Panel A: First child, partner takes leave (W^1)</i>					
-3	-0.138	-0.146	0.008	0.025	[-0.040, 0.056]
3	-0.856	-0.832	-0.024	0.025	[-0.074, 0.025]
9	-0.362	-0.597	0.234***	0.034	[0.168, 0.301]
18	-0.054	-0.037	-0.017	0.040	[-0.095, 0.062]
36	-0.034	-0.012	-0.022	0.057	[-0.133, 0.090]
<i>Panel B: First child, partner does not take leave (W^0)</i>					
-3	-0.150	-0.232	0.081*	0.043	[-0.003, 0.166]
3	-0.893	-0.830	-0.062	0.046	[-0.152, 0.027]
9	-0.598	-0.740	0.142**	0.058	[0.028, 0.256]
18	-0.113	-0.082	-0.031	0.074	[-0.176, 0.115]
36	-0.063	-0.062	-0.001	0.103	[-0.204, 0.201]
<i>Panel C: Second child, partner takes leave (W^1)</i>					
-3	-0.092	-0.215	0.123***	0.021	[0.081, 0.165]
3	-0.824	-0.833	0.009	0.020	[-0.031, 0.049]
9	-0.327	-0.530	0.203***	0.027	[0.150, 0.256]
18	0.012	-0.042	0.054*	0.029	[-0.003, 0.111]
36	0.053	-0.026	0.079**	0.039	[0.003, 0.155]
<i>Panel D: Second child, partner does not take leave (W^0)</i>					
-3	-0.105	-0.271	0.166***	0.027	[0.113, 0.220]
3	-0.866	-0.803	-0.063**	0.025	[-0.111, -0.014]
9	-0.592	-0.742	0.150***	0.031	[0.090, 0.210]
18	-0.038	-0.068	0.030	0.035	[-0.039, 0.099]
36	0.020	-0.058	0.078*	0.046	[-0.012, 0.167]

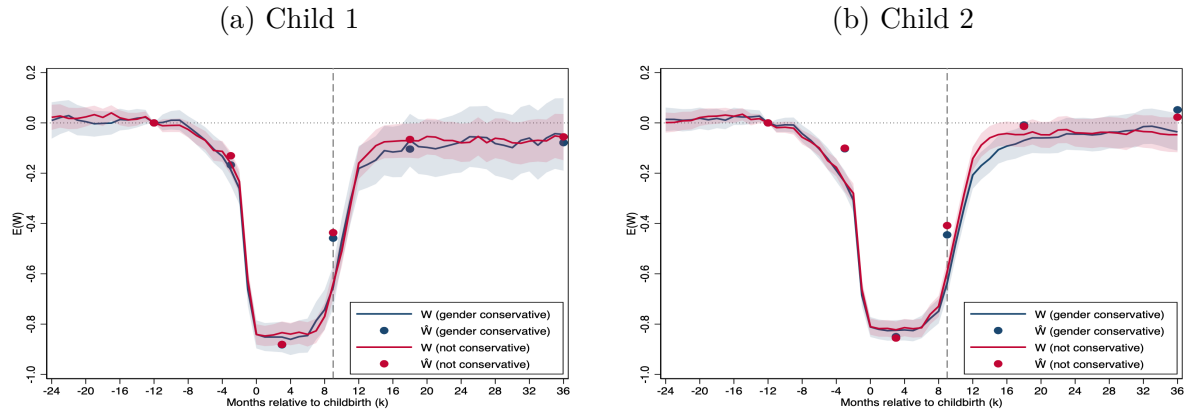
Notes: k denotes months relative to childbirth. Survey and Admin. (pooled) columns report the survey-elicited and pooled TWFE event-study estimates of state-contingent employment (conditional on the partner's realized/expected leave-taking), expressed as deviations from $k = -12$. Admin. estimates include age and calendar year-month fixed effects (equation (1)). Gap = Survey - Admin. $SE(\text{Gap}) = \sqrt{SE(\text{Survey})^2 + SE(\text{Admin})^2}$, which assumes the two estimates are independent; since both are estimated on the same women, this is a conservative (slightly too wide) approximation rather than the exact joint SE. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, two-sided z -test of Gap = 0.

Figure A4: Expected and realized employment by 1st and 2nd child by level of earnings



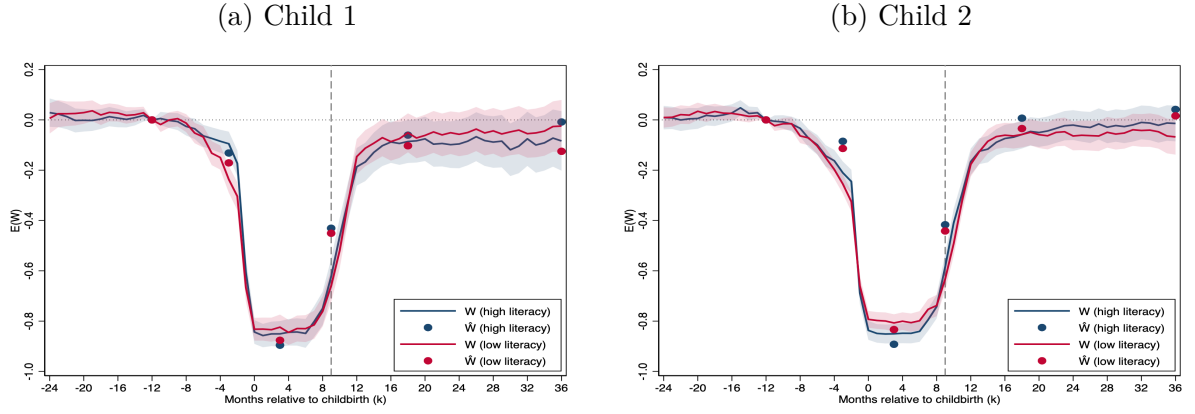
Notes: This figure shows average actual employment, W_i , around childbirth estimated according to (1) and estimated separately for mothers with above median earnings (blue) and below median earnings (red). Median earnings of first/second-time mothers is 300k/330k DKK. Solid lines are estimated actual employment and dots are average unconditional expected employment, $\widehat{W}_i = \hat{p}_i \times \widehat{W}_i^1 + (1 - \hat{p}_i) \times \widehat{W}_i^0$. The shaded region shows the 95% confidence interval for realized employment, W_i ; confidence intervals for expected employment, $\widehat{W}_i$, are omitted from the figure for visual clarity. Panel (a) includes first-time mothers. Panel (b) includes second-time mothers.

Figure A5: Expected and realized employment by 1st and 2nd child by gender conservativeness



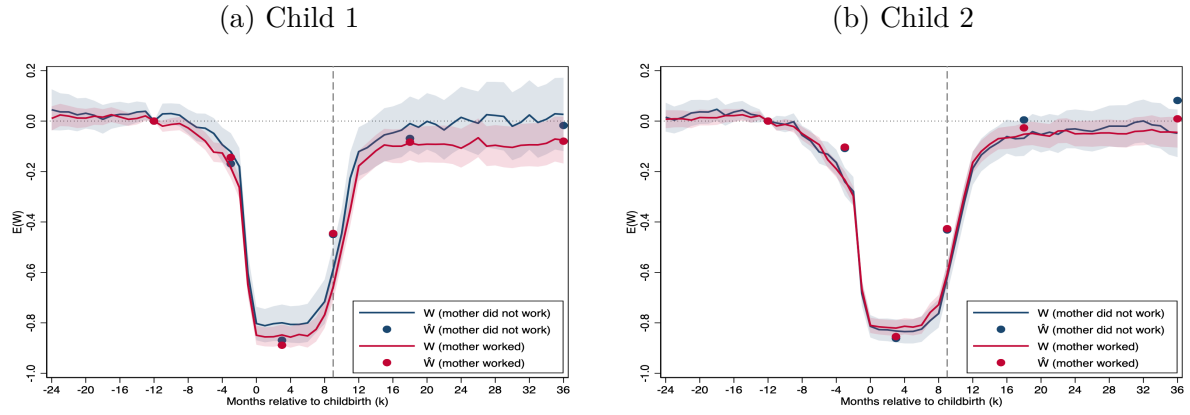
Notes: This figure shows average actual employment, W_i , around childbirth estimated according to (1) and estimated separately by whether the mother is gender conservative (blue) or not (red). Gender conservativeness is defined from answers to questions listed in Appendix B.2. Solid lines are estimated actual employment and dots are average unconditional expected employment, $\widehat{W}_i = \hat{p}_i \times \widehat{W}_i^1 + (1 - \hat{p}_i) \times \widehat{W}_i^0$. The shaded region shows the 95% confidence interval for realized employment, W_i ; confidence intervals for expected employment, $\widehat{W}_i$, are omitted from the figure for visual clarity. Panel (a) includes first-time mothers. Panel (b) includes second-time mothers.

Figure A6: Expected and realized employment by 1st and 2nd child by financial literacy



Notes: This figure shows average actual employment, W_i , around childbirth estimated according to (1) and estimated separately for mothers with high (blue) and low (red) scores in the financial literacy test. Solid lines are estimated actual employment and dots are average unconditional expected employment, $\hat{W}_i = \hat{p}_i \times \hat{W}_i^1 + (1 - \hat{p}_i) \times \hat{W}_i^0$. The shaded region shows the 95% confidence interval for realized employment, W_i ; confidence intervals for expected employment, $\hat{W}_i$, are omitted from the figure for visual clarity. Panel (a) includes first-time mothers. Panel (b) includes second-time mothers.

Figure A7: Expected and realized employment by 1st and 2nd child by whether mother of respondent worked when she was a child



Notes: This figure shows average actual employment, W_i , around childbirth estimated according to (1) and estimated separately for respondents whose own mother did not work (blue) and worked (red) when the respondent was a child. Solid lines are estimated actual employment and dots are average unconditional expected employment, $\widehat{W}_i = \hat{p}_i \times \widehat{W}_i^1 + (1 - \hat{p}_i) \times \widehat{W}_i^0$. The shaded region shows the 95% confidence interval for realized employment, W_i ; confidence intervals for expected employment, $\widehat{W}_i$, are omitted from the figure for visual clarity. Panel (a) includes first-time mothers. Panel (b) includes second-time mothers.

Table A5: Formal Test of Heterogeneity: Realized-Employment Gap

Dimension	Child	$k = -3$	$k = 3$	$k = 9$	$k = 18$	$k = 36$
Education (college)	Child 1	0.099*** (0.038)	-0.085** (0.037)	0.060 (0.049)	0.042 (0.060)	0.031 (0.083)
Education (college)	Child 2	0.067** (0.030)	-0.103*** (0.026)	0.049 (0.035)	0.003 (0.039)	-0.004 (0.052)
Age (≥ 31)	Child 1	0.059 (0.037)	-0.086** (0.035)	0.074 (0.047)	0.013 (0.057)	0.007 (0.081)
Age (≥ 31)	Child 2	0.095*** (0.032)	-0.052* (0.028)	0.085** (0.038)	0.097** (0.049)	0.110 (0.075)
Earnings (above median, 300k)	Child 1	0.055 (0.037)	-0.002 (0.036)	0.028 (0.048)	0.017 (0.057)	0.068 (0.079)
Earnings (above median, 330k)	Child 2	0.080*** (0.031)	0.044* (0.026)	0.097*** (0.035)	0.029 (0.039)	0.063 (0.051)
Gender conservative	Child 1	-0.026 (0.039)	-0.018 (0.038)	-0.012 (0.052)	0.002 (0.064)	0.011 (0.087)
Gender conservative	Child 2	0.000 (0.030)	-0.002 (0.027)	-0.044 (0.036)	-0.023 (0.039)	0.011 (0.052)
Financial literacy (high)	Child 1	0.142*** (0.036)	-0.027 (0.036)	0.037 (0.049)	-0.034 (0.059)	-0.059 (0.080)
Financial literacy (high)	Child 2	0.044 (0.030)	-0.043* (0.026)	0.048 (0.035)	0.000 (0.039)	0.054 (0.051)
Mother did not work	Child 1	0.063 (0.043)	0.048 (0.043)	0.068 (0.056)	0.076 (0.065)	0.102 (0.088)
Mother did not work	Child 2	-0.007 (0.033)	-0.011 (0.028)	-0.012 (0.038)	-0.016 (0.043)	-0.004 (0.057)

Notes: "High" and "low" as defined per dimension (e.g. college vs. non-college). Tests whether realized employment differs by subgroup at each surveyed horizon. Underlying regressions include age and calendar year-month fixed effects (equation (1) in the main text). $k = -12$ is omitted: it is the reference category in the underlying event-study regressions, so the gap is 0 there by construction for every subgroup. Both subgroups in each split are non-overlapping subsamples, so SEs are combined as $\sqrt{se_1^2 + se_2^2}$ exactly, no clustering across groups required. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Formal Test of Heterogeneity: Belief-Error Gap

Dimension	Child	$k = -3$	$k = 3$	$k = 9$	$k = 18$	$k = 36$
Education (college)	Child 1	-0.058 (0.046)	0.066 (0.046)	0.026 (0.062)	0.032 (0.074)	0.038 (0.106)
Education (college)	Child 2	-0.058 (0.036)	0.058* (0.035)	-0.018 (0.046)	0.020 (0.049)	0.018 (0.065)
Age (≥ 31)	Child 1	-0.080* (0.048)	0.064 (0.053)	-0.095 (0.067)	-0.029 (0.078)	-0.026 (0.108)
Age (≥ 31)	Child 2	-0.006 (0.039)	-0.005 (0.038)	-0.069 (0.051)	-0.039 (0.061)	-0.149 (0.091)
Earnings (above median, 300k)	Child 1	-0.035 (0.045)	-0.014 (0.044)	0.057 (0.060)	0.039 (0.071)	-0.015 (0.103)
Earnings (above median, 330k)	Child 2	-0.058 (0.037)	-0.048 (0.035)	-0.054 (0.045)	-0.011 (0.048)	-0.045 (0.063)
Gender conservative	Child 1	-0.011 (0.047)	0.016 (0.047)	-0.010 (0.064)	-0.040 (0.079)	-0.034 (0.112)
Gender conservative	Child 2	-0.003 (0.036)	0.008 (0.035)	0.007 (0.046)	0.029 (0.049)	0.018 (0.064)
Financial literacy (high)	Child 1	-0.103** (0.044)	0.007 (0.045)	-0.017 (0.061)	0.077 (0.073)	0.176* (0.104)
Financial literacy (high)	Child 2	-0.016 (0.036)	-0.015 (0.034)	-0.023 (0.045)	0.041 (0.048)	-0.028 (0.063)
Mother did not work	Child 1	-0.088* (0.053)	-0.028 (0.052)	-0.073 (0.068)	-0.062 (0.079)	-0.039 (0.111)
Mother did not work	Child 2	0.002 (0.040)	0.004 (0.038)	0.007 (0.050)	0.048 (0.053)	0.076 (0.070)

Notes: "High" and "low" as defined per dimension (e.g. college vs. non-college). Tests whether the (survey – admin) over-estimation itself differs by subgroup at each surveyed horizon – the quantity relevant to the paper’s heterogeneity claim. Underlying admin. regressions include age and calendar year-month fixed effects (equation (1) in the main text). Each subgroup’s survey-admin gap, $\widehat{W}_g - W_g$, is the difference of two estimated quantities; its standard error, $\sqrt{se_{\widehat{W}_g}^2 + se_{W_g}^2}$, treats the survey and admin estimates as independent, which is conservative since both are measured on the same women – any positive covariance between stated and realized outcomes (i.e., informative beliefs) would shrink the true SE below this value. $k = -12$ is omitted: it is the reference category in the underlying event-study regressions, so the gap is 0 there by construction for every subgroup. Both subgroups in each split are non-overlapping subsamples, so the two within-group gaps are then combined across groups as $\sqrt{se_1^2 + se_2^2}$ exactly, with no analogous independence assumption needed at this step. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Summary Statistics for Terciles of $\hat{p}$

	Child 1			Child 2		
	Low	Medium	High	Low	Medium	High
Age	27.914	27.382	28.156	29.679	29.452	30.325
Earnings (1000 DKK)	273.767	273.695	294.175	292.237	315.315	353.031
College	0.454	0.579	0.585	0.449	0.565	0.669
Gender Conservative	0.408	0.355	0.363	0.442	0.437	0.398
Financial Literacy	1.980	2.114	2.283	1.881	1.966	2.320

C.4 Robustness to Heterogeneous Treatment Effects: A Cohort-Aggregated Estimator

Equation (1) pools women who give birth in different calendar months into a single two-way-fixed-effects (TWFE) regression. With staggered treatment timing, such pooled estimates can be biased if treatment effects are heterogeneous across cohorts, since already-treated cohorts can act as (implicitly negatively weighted) controls for later cohorts (Sun and Abraham, 2021). This appendix implements a cohort-aggregated event-study estimator in the spirit of Sun and Abraham (2021) that avoids this contamination by construction, and compares its estimates to the pooled specification used in the main text.

Estimator

Let e index treatment cohorts – here, the year-month in which a woman gives birth – and let k denote event time, months relative to that birth. We first estimate every cohort-specific event-time effect, $CATT_{e,k}$, jointly in a single saturated regression:

$$W_{it} = \sum_e \sum_k \mathbf{I}[E_i = e] \cdot \mathbf{I}[t - E_i = k] \cdot \delta_{e,k} + X'_{it}\gamma + u_{it},$$

where E_i is individual i 's treatment cohort and X_{it} is the same set of controls as in Equation (1). Because the regressors are a full set of cohort $\times$ event-time interaction dummies, each coefficient $\delta_{e,k}$ is, by construction, the mean outcome at event time k for cohort e relative to its own baseline period – i.e. $\delta_{e,k} \equiv CATT_{e,k}$, the cohort-specific average treatment effect on the treated. Stacking all N estimated cohort-event-time coefficients into a vector b ($N \times 1$) and their joint sampling covariance into $V = \text{Var}(b)$ ($N \times N$), this single regression delivers every $CATT_{e,k}$ together with its full sampling covariance.

The aggregate dynamic treatment effect at event time k is a weighted average of the cohort-specific effects,

$$ATT_k = \sum_e w_e CATT_{e,k},$$

where w_e is cohort e 's share of unique individuals in the estimation sample. This is a linear combination of b : define a row vector a_k of length N that places weight w_e in the position of $CATT_{e,k}$ and zero elsewhere, so that $ATT_k = a_k b$. Stacking a_k for every event time $k = k_1, \dots, k_K$ in the study window gives a matrix A ($K \times N$), and the full vector of dynamic treatment effects is

$$\theta \equiv \begin{bmatrix} ATT_{k_1} \\ \vdots \\ ATT_{k_K} \end{bmatrix} = A b.$$

Because θ is a linear transformation of b , its exact sampling variance follows directly from the delta method:

$$\text{Var}(\theta) = A V A', \quad SE(ATT_k) = \sqrt{(A V A')_{kk}}.$$

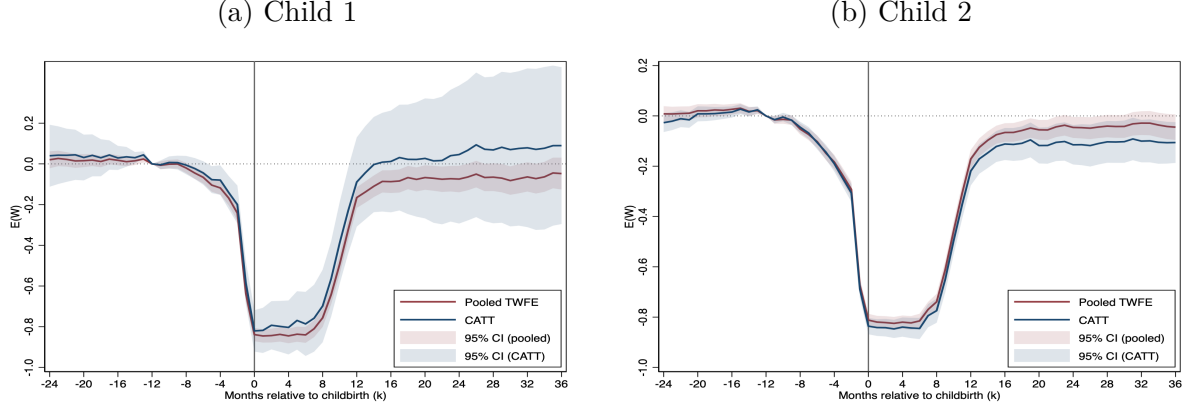
This uses the full covariance structure already estimated in the single saturated regression, requires no bootstrap, and delivers exact, analytically correct standard errors for the aggregated event-study coefficients.

Application

We implement this estimator for the same realized-employment outcome and sample underlying Figure 1, defining treatment cohorts e by the woman's year-month of childbirth (47 cohorts, covering births from 2019m6 to 2023m4) and event time k exactly as in Equation (1), controlling for the same set of age, calendar year-month, and adjacent-child-timing dummies. Cohort weights w_e are each cohort's share of unique mothers in the estimation sample. Figure A8 compares the resulting cohort-aggregated event-study, ATT_k (navy), to the pooled TWFE estimates reported in the main text (maroon), separately for first- and second-time mothers.

The two estimators trace out closely comparable dynamic profiles: both show employment falling sharply around childbirth and recovering over the following year, with the pre-birth trends and the depth and timing of the trough essentially indistinguishable between the two approaches. The cohort-aggregated confidence intervals are visibly wider than the pooled intervals, reflecting the much larger number of parameters estimated

Figure A8: Realized Employment: Pooled TWFE versus Cohort-Aggregated (CATT) Event Study



Notes: This figure compares the pooled two-way-fixed-effects event-study estimates used in the main text (maroon) to the cohort-aggregated estimator described above (navy), both expressed as deviations from $k = -12$. Shaded regions show 95% confidence intervals for each estimator. Panel (a) includes first-time mothers. Panel (b) includes second-time mothers.

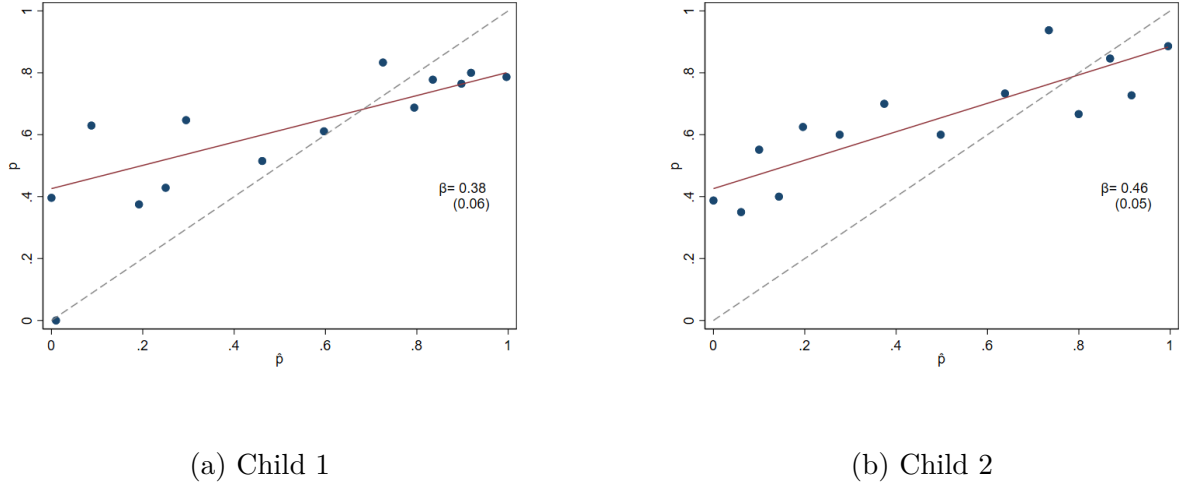
in the saturated cohort-by-event-time regression prior to aggregation; the two sets of confidence intervals overlap throughout the horizon. In the post-birth recovery period the two estimators diverge modestly in level – the cohort-aggregated estimate for first-time mothers returns closer to its pre-birth baseline than the pooled estimate, while for second-time mothers it settles slightly below the pooled estimate – but neither pattern alters the substantive conclusions drawn from Figure 1 in the main text. We conclude that our pooled TWFE estimates are not an artifact of heterogeneous treatment-effect timing across birth cohorts.

C.5 Robustness to Restricting the Sample to Mothers Pregnant at the Survey

A woman’s beliefs about her partner’s leave-taking and her own return to work could change between the 2019 survey and conception – for example, if fertility decisions and beliefs about the division of caregiving are jointly determined. To assess this, we replicate our two central empirical exhibits on the subsample of mothers who were already pregnant at the time of the survey, so that beliefs and the eventual pregnancy are contemporaneous.

For the Figure 3-style comparison, we have already reported the headline result in Section 4.3: among mothers who were pregnant at the time of the survey, the correlation between expected and realized partner leave is somewhat higher than in the full sample – 0.38 (SE = 0.06) for first-time mothers and 0.46 (SE = 0.05) for second-time mothers, compared to 0.26 and 0.39 in the full sample. The underlying figures are reproduced below.

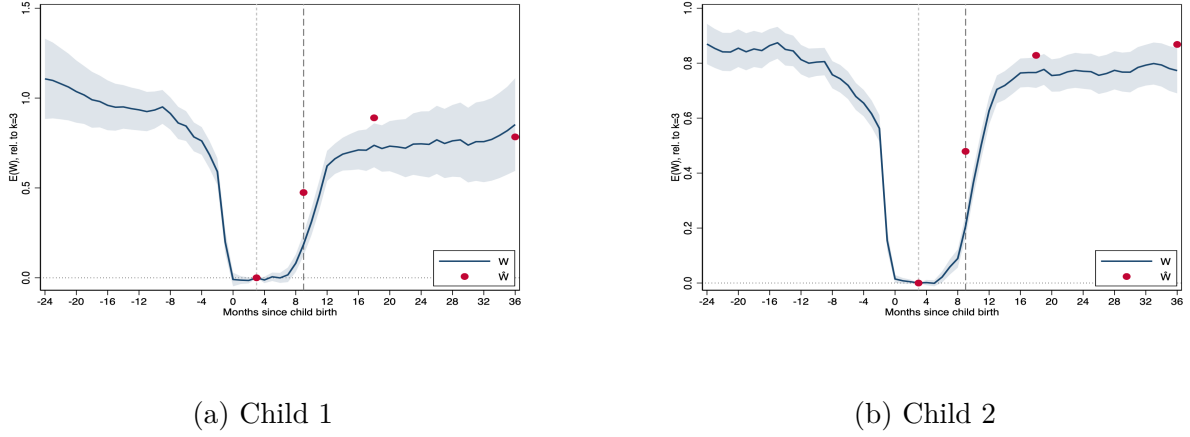
Figure A9: Realized partner leave by stated expectation, pregnant-at-survey subsample



Notes: Binned scatter plot of the fraction of partners taking parental leave, p , against the expectation of the mother that the partner will take leave, $\hat{p}$, restricted to mothers pregnant at the time of the survey, with the fitted regression line (solid) and the 45-degree line of perfect calibration (dashed) shown for reference. Panel (a) includes 360 first-time mothers; panel (b) includes 478 second-time mothers.

For the Figure 1-style comparison, one complication is worth flagging: mothers who were already pregnant at the time of the survey were not asked about expected labor supply at $k = -12$ and $k = -3$, since those pre-birth horizons had already passed or were imminent for an already-pregnant respondent. We therefore re-estimate both the survey and administrative event-study regressions for this subsample with the omitted reference period changed from $k = -12$ to $k = 3$ (the earliest horizon available in both data sources for this group), so the two series remain directly comparable. This necessarily changes what the resulting figure shows – it no longer displays the pre-birth employment decline, since that is absorbed into the (unobserved, for this subsample) $-12 \rightarrow 3$ transition – but it preserves exactly the part of the trajectory most relevant to this robustness check: whether the nine-month overestimation still appears among mothers whose leave planning was more concrete.

Figure A10: Expected and realized employment, pregnant-at-survey subsample (deviations from $k = 3$)



Notes: This figure shows actual employment, W_i , around childbirth (solid line) and unconditional expected employment, $\bar{W}_i$ (dots), for the subsample of mothers who were pregnant at the time of the survey, expressed as deviations from $k = 3$ rather than $k = -12$ (see main text for why). The shaded region shows the 95% confidence interval for realized employment; confidence intervals for expected employment are omitted for visual clarity. The thin dashed line marks $k = 3$, the new reference period; the thick dashed line marks nine months after childbirth ($k = 9$). Panel (a) includes first-time mothers. Panel (b) includes second-time mothers.

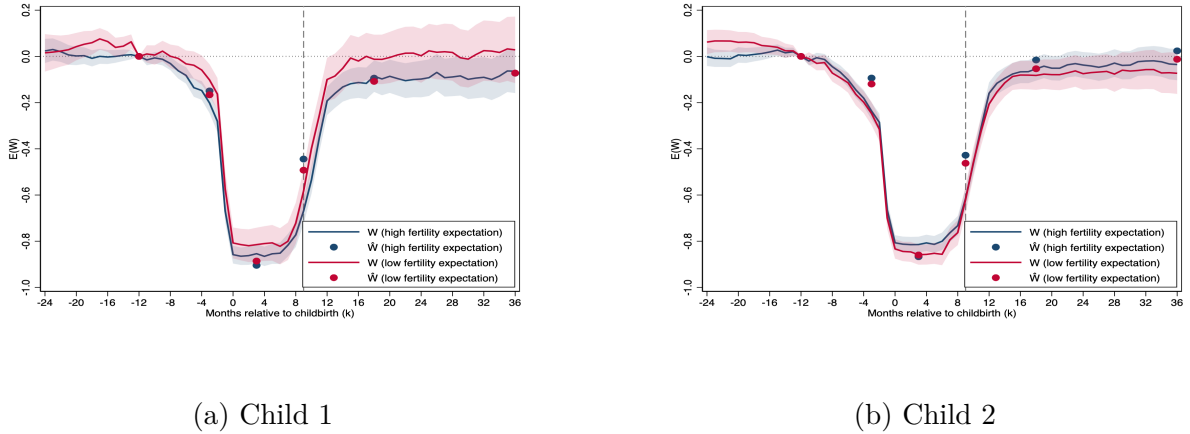
As shown in Figure A10, the core finding replicates. At nine months postpartum, mothers in this subsample still substantially overestimate their return to work, by 0.289 (SE = 0.061) for first-time mothers and 0.273 (SE = 0.032) for second-time mothers, both significant at the 1% level – if anything, these gaps are somewhat larger than in the full sample (0.199 and 0.183, respectively; Appendix Table A3), though the smaller size of this subsample means correspondingly wider standard errors. At 18 and 36 months the gaps are smaller and not statistically significant for first-time mothers (0.153, SE = 0.128; –0.069, SE = 0.292); for second-time mothers the 18-month gap is likewise insignificant (0.062, SE = 0.040) while the 36-month gap is marginally significant at the 10% level (0.095, SE = 0.056). No comparable pre-birth ($k = -12/-3$) horizons are available for this subsample, for the reason noted above. Overall, the central result of the paper – systematic overestimation of return-to-work specifically at nine months, not at other horizons – holds up among mothers whose leave planning was plausibly most concrete.

C.6 Robustness to Splitting by Fertility Expectations

Because our sample by construction consists of women who went on to have a(nother) child, a further concern is selection into birth: whether belief accuracy differs between women who were, at the time of the survey, more or less certain they would go on to have a(nother) child. Briggs et al. (2023) elicit exactly this measure – $\hat{p}_{child}$, the stated probability of having a(nother) child – allowing us to test this directly.

We replicate the Figure 1-style comparison splitting the sample by $\hat{p}_{child}$. One caveat is worth flagging upfront: because our sample by construction consists of women who went on to have a(nother) child, fertility expectations are generally high in this sample rather than spanning the full $[0, 1]$ range – the median split is at $\hat{p}_{child} = 0.85$ for first-time mothers and $\hat{p}_{child} = 0.90$ for second-time mothers. The comparison below is therefore between mothers with very high and moderately high (not low) fertility expectations.

Figure A11: Expected and realized employment by fertility expectation ($\hat{p}_{child}$, above vs. below median)



Notes: This figure shows average actual employment, W_i , around childbirth estimated according to equation (1) and estimated separately for mothers with above-median (blue) and below-median (red) stated probability of having a(nother) child, $\hat{p}_{child}$ (median splits: 0.85 for first-time mothers, 0.90 for second-time mothers; see main text). Solid lines are estimated actual employment; dots are average unconditional expected employment, $\widehat{W}_i$. The shaded region shows the 95% confidence interval for realized employment; confidence intervals for expected employment are omitted for visual clarity. Panel (a) includes first-time mothers. Panel (b) includes second-time mothers.

Figure A11 shows that the two fertility-expectation subgroups follow visually similar trajectories for both parities, with the central nine-month overestimation present in both. Formally testing whether the belief-error gap differs between subgroups, two of ten com-

parisons (5 horizons $\times$ 2 parities) reach conventional significance, both for first-time mothers: at nine months (0.134, SE = 0.069, $p = 0.051$) and three months before birth (0.116, SE = 0.049, $p = 0.018$); second-time mothers show no significant difference at any horizon (all $p > 0.11$). Even these are modest next to the belief errors both subgroups display: at nine months, mothers with above- and below-median fertility expectations overestimate their return to work by 0.225 and 0.091, respectively – both sizeable, both consistent with the paper’s central finding, and differing in degree rather than in kind. We do not think this points to fertility intentions as a confound: the pattern is not replicated for second-time mothers, and two significant results out of ten tests is unremarkable at conventional thresholds.

D Belief Error Decomposition

This appendix derives the decomposition of belief errors presented in Section 4.4. The goal is to separate total belief misalignment into distinct components: errors in state-contingent beliefs, errors in state beliefs, and residual sorting terms.

Individual-Level Framework

We define the following notation for each individual i :

- $D_i \in \{0, 1\}$: indicator for whether the partner took leave.
- $p_i = \mathbb{E}[D_i]$: true probability that the partner will take leave.
- $\hat{p}_i$: stated (subjective) probability that the partner will take leave.
- W_i^1 : latent probability of working if the partner takes leave.
- W_i^0 : latent probability of working if the partner does not take leave.
- $\widehat{W}_i^1$: stated probability of working if the partner takes leave.
- $\widehat{W}_i^0$: stated probability of working if the partner does not take leave.

The realized outcome is:

$$W_i = D_i W_i^1 + (1 - D_i) W_i^0.$$

The belief is:

$$\widehat{W}_i = \hat{p}_i \widehat{W}_i^1 + (1 - \hat{p}_i) \widehat{W}_i^0.$$

Taking expectations and applying the identity $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y] + \text{Cov}(X, Y)$, we obtain:

$$\begin{aligned} \mathbb{E}[W_i] &= p_i \mathbb{E}[W_i^1] + (1 - p_i) \mathbb{E}[W_i^0] + \text{Cov}(D_i, W_i^1 - W_i^0), \\ \mathbb{E}[\widehat{W}_i] &= \mathbb{E}[\widehat{W}_i^0] + \mathbb{E}[\hat{p}_i] \mathbb{E}[\widehat{W}_i^1 - \widehat{W}_i^0] + \text{Cov}(\hat{p}_i, \widehat{W}_i^1 - \widehat{W}_i^0). \end{aligned}$$

Thus, the belief error is:

$$\begin{aligned}
\text{Bias}_i &= \mathbb{E}[W_i] - \mathbb{E}[\widehat{W}_i] \\
&= (1 - p_i)(\mathbb{E}[W_i^0] - \mathbb{E}[\widehat{W}_i^0]) + p_i(\mathbb{E}[W_i^1] - \mathbb{E}[\widehat{W}_i^1]) \\
&\quad + (p_i - \mathbb{E}[\hat{p}_i])(\mathbb{E}[\widehat{W}_i^1] - \mathbb{E}[\widehat{W}_i^0]) \\
&\quad + \text{Cov}(D_i, W_i^1 - W_i^0) - \text{Cov}(\hat{p}_i, \widehat{W}_i^1 - \widehat{W}_i^0).
\end{aligned}$$

The first two terms capture miscalibration in beliefs about employment within each state. The third term reflects miscalibration in beliefs about the partner's behavior (state belief error), and the final two covariance terms capture residual sorting:

- $\text{Cov}(D_i, W_i^1 - W_i^0)$ reflects whether treatment assignment (partner leave) is correlated with employment treatment effects.
- $\text{Cov}(\hat{p}_i, \widehat{W}_i^1 - \widehat{W}_i^0)$ reflects whether stated expectations about partner leave correlate with expected gains from partner leave.

These terms are not directly estimable because only one of (W_i^1, W_i^0) is observed for each individual.

Group-Level Aggregation

To facilitate empirical implementation, we aggregate the individual-level decomposition to the group level. Let g index a group (e.g., defined by terciles of $\mathbb{E}[\hat{p}_i]$ or the entire sample), and let $\mathbb{E}_g[\cdot]$ denote the expectation taken over individuals $i \in g$. Let $\text{Cov}_g(\cdot, \cdot)$ denote the corresponding within-group covariance.

We define the following group-level quantities:

- $p_g = \mathbb{E}_g[D_i]$: realized fraction of partners taking leave in group g .
- $\mathbb{E}_g[\hat{p}_i]$: average subjective probability that the partner will take leave in group g .
- $\mathbb{E}_g[W_i^1]$: average probability of working if the partner takes leave.
- $\mathbb{E}_g[W_i^0]$: average probability of working if the partner does not take leave.
- $\mathbb{E}_g[\widehat{W}_i^1]$: average stated probability of working if the partner takes leave.
- $\mathbb{E}_g[\widehat{W}_i^0]$: average stated probability of working if the partner does not take leave.

The group-level realized outcome is:

$$\mathbb{E}_g[W_i] = p_g \mathbb{E}_g[W_i^1] + (1 - p_g) \mathbb{E}_g[W_i^0] + \text{Cov}_g(D_i, W_i^1 - W_i^0),$$

and the group-level belief is:

$$\mathbb{E}_g[\widehat{W}_i] = \mathbb{E}_g[\widehat{W}_i^0] + \mathbb{E}_g[\widehat{p}_i] \mathbb{E}_g[\widehat{W}_i^1 - \widehat{W}_i^0] + \text{Cov}_g(\widehat{p}_i, \widehat{W}_i^1 - \widehat{W}_i^0).$$

The resulting group-level bias is:

$$\begin{aligned} \text{Bias}_g &= \mathbb{E}_g[W_i] - \mathbb{E}_g[\widehat{W}_i] \\ &= (1 - p_g)(\mathbb{E}_g[W_i^0] - \mathbb{E}_g[\widehat{W}_i^0]) + p_g(\mathbb{E}_g[W_i^1] - \mathbb{E}_g[\widehat{W}_i^1]) \\ &\quad + (p_g - \mathbb{E}_g[\widehat{p}_i])(\mathbb{E}_g[\widehat{W}_i^1] - \mathbb{E}_g[\widehat{W}_i^0]) \\ &\quad + \text{Cov}_g(D_i, W_i^1 - W_i^0) - \text{Cov}_g(\widehat{p}_i, \widehat{W}_i^1 - \widehat{W}_i^0). \end{aligned}$$

Each term has a natural interpretation:

- The first two terms reflect average miscalibration in state-contingent beliefs.
- The third line captures average miscalibration in state beliefs about partner leave.
- The final line includes sorting terms, which capture how partner leave is correlated with realized or expected treatment effects within group g .

The first sorting term is not directly estimable, since individual-level counterfactuals (W_i^1, W_i^0) cannot be jointly observed. However, given that all the other terms are observable, this term can be derived residually.

D.1 Components of the Decomposition

Table A8 reports the individual levels – p_g , $\mathbb{E}_g[\widehat{p}_i]$, $\mathbb{E}_g[W_i^1]$, $\mathbb{E}_g[W_i^0]$, $\mathbb{E}_g[\widehat{W}_i^1]$, $\mathbb{E}_g[\widehat{W}_i^0]$, and the two residual covariance terms – underlying the decomposition reported in Table 1.

D.2 Robustness to the August 2022 Parental Leave Reform

Parental leave rules changed in August 2022, reserving 11 additional weeks (about 2.5 months) of the shared 32 weeks specifically for fathers (Section 2, footnote 3). Because our survey elicited beliefs in 2019 under the pre-reform rules, and because we define partner leave-taking throughout the paper as taking *at least two months*, a mechanical concern is

Table A8: Components of the decomposition at $k = 9$, by levels of $\hat{p}$:

	Child 1			Child 2		
	Low	Medium	High	Low	Medium	High
p_g	0.596*** (0.023)	0.736*** (0.020)	0.751*** (0.017)	0.454*** (0.023)	0.688*** (0.021)	0.748*** (0.018)
$\mathbb{E}_g[\hat{p}]$	0.174*** (0.007)	0.671*** (0.006)	0.959*** (0.002)	0.093*** (0.004)	0.600*** (0.008)	0.967*** (0.002)
$\mathbb{E}_g[W_i^1]$	0.250*** (0.047)	0.370*** (0.046)	0.503*** (0.048)	0.269*** (0.041)	0.411*** (0.041)	0.615*** (0.034)
$\mathbb{E}_g[W_i^0]$	0.099 (0.065)	0.333** (0.131)	0.386*** (0.103)	0.208*** (0.039)	0.244*** (0.056)	0.331*** (0.051)
$\mathbb{E}_g[\widehat{W}_i^1]$	0.624*** (0.042)	0.649*** (0.029)	0.624*** (0.028)	0.643*** (0.031)	0.669*** (0.025)	0.691*** (0.025)
$\mathbb{E}_g[\widehat{W}_i^0]$	0.409*** (0.043)	0.459*** (0.036)	0.356*** (0.028)	0.427*** (0.027)	0.447*** (0.026)	0.330*** (0.025)
$\text{cov}(D, W^1 - W^0)$	0.004 (0.077)	-0.005 (0.075)	-0.003 (0.071)	0.003 (0.054)	-0.005 (0.056)	0.023 (0.052)
$\text{cov}(\hat{p}, \widehat{W}^1 - \widehat{W}^0)$	-0.002 (0.002)	-0.000 (0.002)	0.002** (0.001)	-0.001 (0.001)	0.002 (0.003)	0.000 (0.001)

Notes: The table presents the components entering the decomposition, cf. equation (2). ‘Low’, ‘Medium’, and ‘High’ indicate the strata defined by the lower, middle and upper tercile of the distribution of $\hat{p}$ (see Table 1 for tercile sample sizes). Underlying regressions for $\mathbb{E}_g[W_i^1]$, $\mathbb{E}_g[W_i^0]$ include age and calendar year-month fixed effects (equation (1)). These are primitive quantities rather than derived combinations, so standard errors in parentheses are obtained directly from the underlying regression and survey estimates rather than via the delta method (contrast Table 1, whose subtotal and total rows combine several of these primitives and therefore do require the delta method). As in Table 1, the residual sorting terms (rows $\text{cov}(D, W^1 - W^0)$ and $\text{cov}(\hat{p}, \widehat{W}^1 - \widehat{W}^0)$) are economically small relative to the other rows even where individually significant. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

that the reform alone could push partners in later cohorts over this threshold, independent of any change in underlying behavior or beliefs. This would be a particular concern for the positive state belief error we document among mothers with low $\hat{p}$ (Table 1): if the reform mechanically increases realized partner leave-taking among late cohorts, part of the gap between p_g and $\mathbb{E}_g[\hat{p}_i]$ in this group could reflect the reform rather than mothers underestimating their partners’ choices under the rules that actually applied when they were surveyed.

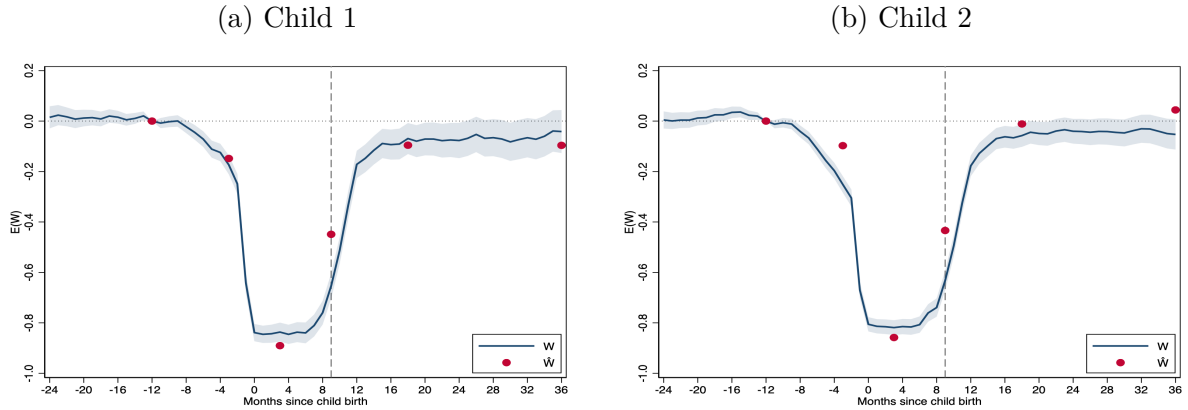
To address this directly, we re-estimate Figures 1 and 2 and the Table 1 decomposition restricting the sample to births before August 2022, so that every mother in this subsample was subject to the same parental leave rules in place at the time of the survey.

Table A9: Formal Test of Learning Between Births: Child 1 vs. Child 2

	Low	Medium	High
Total bias	-0.045 (0.068)	-0.003 (0.066)	-0.029 (0.061)
State-contingent error	-0.059 (0.060)	0.002 (0.066)	-0.026 (0.060)
State belief error	0.013 (0.030)	-0.007 (0.008)	0.023* (0.014)

Notes: Each cell reports the difference (Child 1 minus Child 2) in the corresponding row of Table 1 at $k = 9$, within the indicated $\hat{p}$ tercile, with its standard error in parentheses below. Both children's underlying estimates have standard errors obtained via the delta method (as in Table 1); the difference's standard error combines them as $\sqrt{se_1^2 + se_2^2}$. This is exact only if the child-1 and child-2 estimation samples are independent; 392 mothers in our data appear in both, so this is an approximation rather than an exact standard error. Of the nine tests reported here, only one – the widening of the state belief error in the high- $\hat{p}$ tercile – is significant at conventional levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A12: Expected and realized employment by 1st and 2nd child, births before August 2022

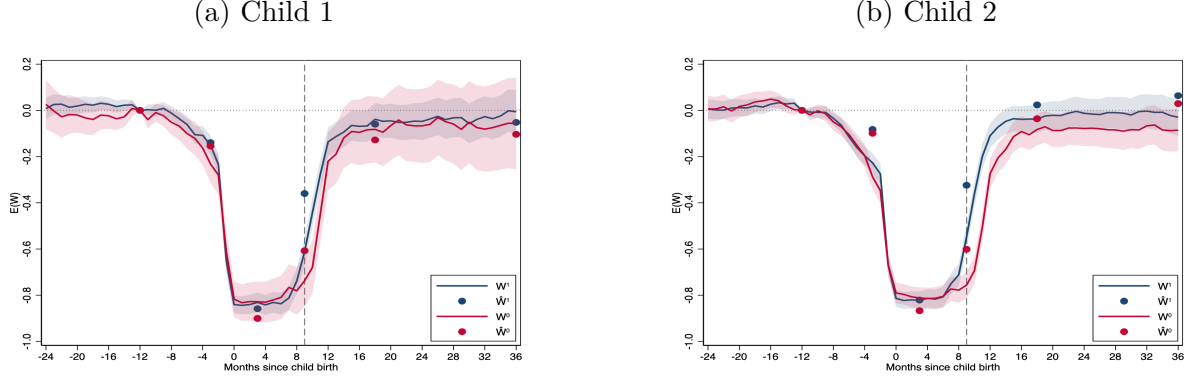


Notes: Same construction as Figure 1, restricted to mothers whose child was born before August 2022, i.e. under the parental leave rules in place at the time of the 2019 survey. Panel (a) includes first-time mothers. Panel (b) includes second-time mothers. The dashed line marks nine months after childbirth ($k = 9$).

Figures A12 and A13 reproduce Figures 1 and 2 on this restricted sample; Table A10 below reproduces the Table 1 decomposition.

The pattern most relevant to the reform concern is the state belief error among mothers with low $\hat{p}$ – the component that a mechanical, reform-driven increase in partner

Figure A13: Expected and Realized Employment by Whether Partner Taking Parental Leave, births before August 2022



Notes: Same construction as Figure 2, restricted to mothers whose child was born before August 2022. Panel (a) includes first-time mothers. Panel (b) includes second-time mothers. The dashed line marks nine months after childbirth ($k = 9$).

leave-taking could inflate. Restricting to births before August 2022 leaves this essentially unchanged: 0.096 (SE= 0.023) for the first child and 0.077 (SE= 0.015) for the second, against 0.091 (SE= 0.026) and 0.078 (SE= 0.016) in the full sample – if anything, marginally larger for the first child once the reform-affected cohorts are excluded entirely, which is inconsistent with the reform being the primary driver of this result. Total bias and the state-contingent component are also similar in sign, magnitude, and significance across the two samples in every tercile, with the high- $\hat{p}$ tercile the only case where a point estimate moves by an economically meaningful amount (Total bias for Child 1 High falls to -0.133 from -0.143 , and loses one star, consistent with the smaller sample rather than a different underlying pattern). This pattern is inconsistent with the reform driving our central results.

Table A10: Decomposition of Expectation Bias at $k = 9$, by Levels of $\hat{p}$: Births Before August 2022

	Low	Child 1 Medium	High	Low	Child 2 Medium	High
(1) State-contingent belief error						
$(1 - p_g)(\mathbb{E}_g[W_i^0] - \mathbb{E}_g[\widehat{W}_i^0])$	-0.124*** (0.035)	-0.038 (0.037)	0.010 (0.031)	-0.133*** (0.028)	-0.070*** (0.025)	0.002 (0.016)
$p_g(\mathbb{E}_g[W_i^1] - \mathbb{E}_g[\widehat{W}_i^1])$	-0.224*** (0.038)	-0.233*** (0.039)	-0.080* (0.044)	-0.163*** (0.026)	-0.186*** (0.042)	-0.074** (0.036)
Subtotal: State-contingent error	-0.348*** (0.050)	-0.270*** (0.054)	-0.070 (0.054)	-0.295*** (0.037)	-0.256*** (0.048)	-0.072* (0.040)
(2) State belief error						
$(p_g - \mathbb{E}_g[\hat{p}_i])(\mathbb{E}_g[\widehat{W}_i^1] - \mathbb{E}_g[\widehat{W}_i^0])$	0.096*** (0.023)	0.007 (0.005)	-0.069*** (0.012)	0.077*** (0.015)	0.003 (0.006)	-0.090*** (0.012)
(3) Residual sorting terms						
$\text{cov}(D, W^1 - W^0)$	0.008 (0.075)	-0.009 (0.074)	0.007 (0.078)	0.002 (0.056)	-0.007 (0.070)	0.030 (0.060)
$\text{cov}(\hat{p}, \widehat{W}^1 - \widehat{W}^0)$	-0.002 (0.002)	-0.000 (0.002)	0.001* (0.001)	-0.002 (0.002)	0.003 (0.002)	0.001 (0.001)
Total bias	-0.242*** (0.054)	-0.273*** (0.051)	-0.133** (0.054)	-0.214*** (0.041)	-0.263*** (0.052)	-0.132*** (0.042)

Notes: Same construction as Table 1, restricted to births before August 2022. Underlying regressions for W^1 , W^0 include age and calendar year-month fixed effects (equation (1)). Standard errors in parentheses; as in Table 1, the state-contingent subtotal and total bias are obtained by the delta method (via `nlcom`) rather than by combining the component standard errors under an independence assumption. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.